MILITARY SPECIFICATION

* MODULES, STANDARD ELECTRONIC

* BUFFER, THREE-STATE, DIGITAL, KEY CODE BBB

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

* 1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the procurement of an electronic three-state buffer module.

1.2 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table II.

* Name	Buffer, three-state
Family	Digital
Key code	BBB
Size	1A
Weight	35 grams maximum
Failure rate	0.67 failures/10^6 hours maximum
* Environment	Class I as specified in MIL-STD-1389
Supply voltage	5.0 Vdc ±10%
Maximum power dissipation	1.5 W (outputs not loaded)
Input voltage	
Low level	-0.3 V minimum, 0.8 V maximum
High level	2.0 V minimum, V_{CC} maximum
Input current (A1-A6, B1-B6, A disable, B disable)	
Low level (V_{IL} = 0.0 V)	-1.73 mA maximum
(V_{IL} = 0.4 V)	-1.6 mA maximum
High level (V_{IH} = 2.4 V)	40 µA maximum
(V_{IH} = 5.5 V)	1.0 mA maximum
Input current (mode control)	
Low level (V_{IL} = 0.0 V)	-5.19 mA maximum
(V_{IL} = 0.4 V)	-4.8 mA maximum
High level (V_{IH} = 2.4 V)	120 µA maximum
(V_{IH} = 5.5 V)	3.0 mA maximum
Output voltage	
Low level (I_{OL} = 32 mA)	0.4 V maximum
High level (I_{OH} = -2.0 mA)	2.4 V minimum

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command, ATTN: ELEX 5043, Department of the Navy, Washington, D.C. 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Output drive current maximum capability	
Low level (V_{OL} = 0.4 V)	32 mA
(V_{OL} = 0.5 V)	40 mA
High level (V_{OH} = 1.8 V)	-40.0 mA
(V_{OH} = 2.4 V)	-2.0 mA
Output leakage current	
V_{OL} = 0.4 V	-40 µA minimum, 40 µA maximum
V_{OH} = 2.4 V	-40 µA minimum, 40 µA maximum
Propagation delay time (C_L = 100 pF)	
Input to output (tp_{LH})	5.0 ns minimum, 30.0 ns maximum
Input to output (tp_{HL})	5.0 ns minimum, 30.0 ns maximum
Propagation delay time (C_L = 1,000 pF)	
Input to output (tp_{LH})	10 ns minimum, 55.0 ns maximum
Input to output (tp_{HL})	10 ns minimum, 55.0 ns maximum
Propagation delay time (C_L = 50 pF)	
Disable to output (T_{lh})	5.0 ns minimum, 40.0 ns maximum
Disable to output (T_{hl})	5.0 ns minimum, 40.0 ns maximum
Disable to output (T_{oh})	5.0 ns minimum, 40.0 ns maximum
Disable to output (T_{ho})	5.0 ns minimum, 40.0 ns maximum

* 1.3 Module function. This module contains three 4-input three-state line drivers. Each driver has either a quad single-ended driver mode, a dual differential driver mode, or a high-impedance output mode.

a. To operate as 12 single-ended line drivers, apply low levels to the output disable inputs (to keep the outputs in the normal low-impedance mode) and apply a low level to the differential/single-ended mode control input. All channels will then operate independently and no output signal inversion will occur.

b. To operate as 6-dual differential line drivers, apply low levels to the output disable inputs and apply a high level to the differential/single-ended mode control input. The A inputs should be tied together and the B inputs should be tied together on each driver circuit. In this mode, the signals applied to the respective inputs will pass noninverted to A2 and B2 outputs and inverted to the A1 and B1 outputs.

c. When operating in a bus-oriented system with outputs tied directly to outputs of similar devices, all devices except one shall be placed in the high-impedance state. This is accomplished by insuring that a high level is applied to the output disable inputs of all devices except the one that is used to drive the bus. The driving device provides good capacitive drive capability and waveform integrity especially on the positive transition

1.3.1 Input to output relationship. Each buffer has the input to output relationship of table I for positive logic.

TABLE I. Input to output relationship.

Inputs							Outputs			
Mode control	A out disable	B out disable	A1	A2	B1	B2	A1	A2	B1	B2
L	L L	X X	$\overline{A1}$	A2	X	X	$\overline{A1}$	A2	X	X
H	L L	X X	A1	A2	X	X	$\overline{A1}$	A2	X	X
X	H X	X X	X	X	X	X	HIS	HIS	X	X
X	X H	X X	X	X	X	X	HIS	HIS	X	X
L	X X	L L	X	X	B1	B2	X	X	$\overline{B1}$	B2
H	X X	L L	X	X	B1	B2	X	X	$\overline{B1}$	B2
X	X X	H X	X	X	X	X	X	X	HIS	HIS
X	X X	X H	X	X	X	X	X	X	HIS	HIS

NOTES H = high-voltage level. HIS = high-impedance state.
L = low-voltage level X = don't care

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1.3.2 Parallel outputs. Paralleling of outputs is permissible provided only one output at a time is in a high or low state and all other outputs are in the high-impedance state.

1.3.3 Module crosstalk. When connected in the worst case single-ended mode of operation (all but one buffer switching simultaneously with the same polarity), the BBB module could exhibit substantial crosstalk. When the quiet (nonswitching) output is at a low level, there could be as much as a 1.49 volts positive going spike coincident with the negative edge of the switching outputs. When the quiet output is at a high level, there could be as much as a 700 mV negative going spike coincident with the positive or negative edge of the switching output.

1.3.4 Single-ended mode crosstalk. The crosstalk on any output pin is nearly directly proportional to the number of simultaneously switching outputs.

a. For every switching output in group A (buffers A1, A2, B1, and B2), the noise induced on a low-level quiet output in group A can be as much as 310 mV. For every switching output outside group A (buffers A3, A4, B3, B4, B5, A6, B5, and B6), the noise induced on a low-level quiet output in group A can be as much as 70 mV).

b. For every switching output in group B (buffers A3, A4, B3, and B4), the noise induced on a low-level quiet output in group B can be as much as 310 mV. For every switching output outside group B (buffers A1, A2, B1, B2, A5, A6, B5, and B6), the noise induced on a low-level output in group B can be as much as 70 mV.

c. For every switching output in group C (buffers A5, A6, B5, and B6), the noise induced on a low-level quiet output in group C can be as much as 310 mV. For every switching output outside group C (buffers A1, A2, B1, B2, A3, A4, B3, and B4), the noise induced on a low-level quiet output in group C can be as much as 70 mV.

1.3.5 Differential mode crosstalk. When connected in the worst case differential mode of operation (all but one differential pair switching simultaneously), the differential crosstalk induced on the quiet differential pair can be as much as 1.0 V.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.
MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

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* 3.3 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 0.67 failures per million operating hours when predicted in accordance with MIL-STD-1389.

* 3.4 Performance requirements. The module shall conform to the electrical requirements specified in table II and shall operate within the environment of the design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be BBB.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Pin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

* 3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/47-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

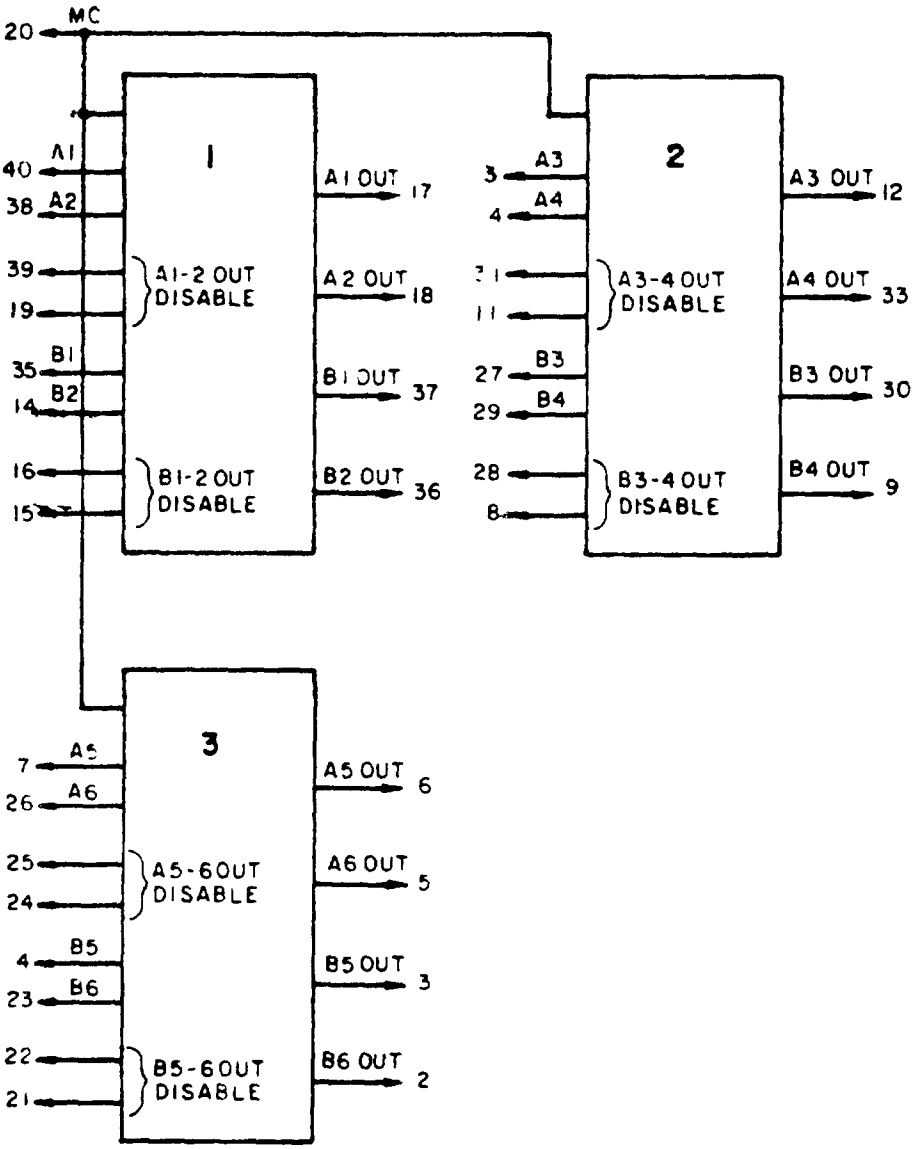
3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance test. All production modules shall be tested in accordance with notes 1 and 2 of table II.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table II (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table II for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc ±1 percent.

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1 ——— V_{CC} (5.0±0.5 Vdc)
10 ——— GND (0V REF)
32 ——— UNUSED

NOTES:
1. MC - mode control input.
2. A1 - A6 = buffer data inputs.
3. B1 - B6 = buffer data inputs.
4. A1-2, A3-4, A5-6 = A out disable inputs.
5. B1-2, B3-4, B5-6 = B out disable inputs.
6. A1 out - A6 out = buffer outputs.
7. B1 out - B6 out = buffer outputs.

FIGURE 1. Module function diagram, three-state buffer

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TABLE II. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A2	Isolation	III	S	10		1		10		1		MΩ
B1-B12	Maximum stress current check	IV	S	-	-	-	-	-	-	-	-	-
C1-C12	V_{OUT} (low-loaded)	V	S	0 01	1.0	0.01	1 0	0 01	1.0	0.01	1 0	V
D1-D24	Input breakdown current (data and disable)	VI	S	0 0	1 0	0.0	1.0	0.0	1.0	0.0	1 0	mA
D25	Input breakdown current (mode control)	VI	S	0 0	3.0	0.0	3.0	0.0	3.0	0.0	3.0	mA
E1-E12	Output short circuit current	VII	X	-40	-120	-40	-120	-40	-120	-40	-120	mA
F1-F25	Input clamp diode voltage	VIII	X	-0 4	-1 5	-0 4	-1.5	-0.4	-1.5	-0.4	-1.5	V
G1-G12	Output clamp diode voltage	VIII	X	-0 4	-1 5	-0 4	-1.5	-0.4	-1.5	-0.4	-1.5	V
H1-H12	Output leakage current, high level	IX	X	-40	40	-40	40	-40	40	-40	40	µA
J1-J12	Output leakage current, power off	IX	X	0	40	0	40	0	40	0	40	µA
K1-K12	Output leakage current, low level	X	X	-40	40	-40	40	-40	40	-40	40	µA
L1-L24	Output voltage, low level, I_{OL} = 32 mA	XI	X	0 0	0.4	0.0	0 4	0.0	0.4	0.0	0.4	V
L25-L36	Output voltage, low level, I_{OL} = 40 mA	XI	X	0.0	0.5	0.0	0.5	0.0	0.5	0.0	0.5	V
M1-M24	Output voltage, high level, I_{OH} = -2 mA	XI	X	2 4	4 5	2 4	4.5	2 4	4.5	2.4	4.5	V
M25-M36	Output voltage, high level, I_{OH} = -40 mA	XI	X	1 8	4.5	1 8	4.5	1 8	4.5	1 8	4.5	V

See footnotes at end of table.

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TABLE II Module electrical requirements - Continued

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
N1-N24	Input current, high level (data and disable)	XII	X	0	40	0	40	0	40	0	40	µA
N25	Input current, high level (mode control)	XII	X	0	120	0	120	0	120	0	120	µA
P1-P24	Input current, low level (data and disable)	XIII	X	-0.1	-1.73	-0.1	-1.73	-0.1	-1.73	-0.1	-1.73	mA
P25	Input current, low level (mode control)	XIII	X	-0.3	-5.19	-0.3	-5.19	-0.3	-5.19	-0.3	-5.19	mA
Q	Power supply current	XIV	X	50	270	50	270	50	270	50	270	mA
R1-R12	Output transition time (t_{TLH}), C_L = 100 pF	XV	S	4	25	4	25	4	28	4	28	ns
S1-S12	Output transition time (t_{THL}), C_L = 100 pF	XV	S	3	20	3	20	3	23	3	23	ns
T1-T18	Propagation delay time (t_{PLH}), C_L = 100 pF	XVI	S	5	25	5	25	5	30	5	30	ns
T19-T36	Propagation delay time (t_{PLH}), C_L = 1,000 pF	XVI	Q	10	50	10	50	10	55	10	55	ns
U1-U18	Propagation delay time (t_{PHL}), C_L = 100 pF	XVI	S	5	25	5	25	5	30	5	30	ns
U19-U36	Propagation delay time (t_{PHL}), C_L = 1,000 pF	XVI	Q	10	50	10	50	10	55	10	55	ns
V1-V24	Disable delay time (toh), CL = 50 pF	XVII	S	5	35	5	35	5	40	5	40	ns

See footnotes at end of table

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TABLE II. Module electrical requirements - Continued

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
W1-W24	Disable delay time (t_{ho}), C_L = 50 pF	XVII	S	5	35	5	35	5	40	5	40	ns
X1-X24	Disable delay time (t_{lh}), C_L = 50 pF	XVII	S	5	35	5	35	5	40	5	40	ns
Y1-Y24	Disable delay time (t_{hl}), C_L = 50 pF	XVII	S	5	35	5	35	5	40	5	40	ns
Z1-Z6	Differential delay time (Δt_{PLH}), C_L = 50 pF	XVIII	S	-10	10	-10	10	-10	10	-10	10	ns
Z7-Z12	Differential (Δt_{PHL}), C_L = 50 pF	XVIII	S	-10	10	-10	10	-10	10	-10	10	ns
AA1-AA12	Output crosstalk, high level (single-ended mode)	XIX	Q	0	700	0	700	0	700	0	700	mV
AA13-AA24	Output crosstalk, high level (single-ended mode)	XIX	Q	0	550	0	550	0	550	0	550	mV
BB1-BB12	Output crosstalk, low level (single-ended mode)	XIX	Q	0.0	1.49	0.0	1.49	0.0	1.49	0.0	1.49	V
BB13-BB24	Output crosstalk, low level (single-ended mode)	XIX	Q	0	560	0	560	0	560	0	560	mV
CC1-CC12	Output crosstalk, (differential mode)	XX	Q	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	V

See footnotes at end of table.

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TABLE II Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature, Initial Min	Initial Max	End-of-life Min	End-of-life Max	Operating temperature 0° to 60°C Initial Min	Initial Max	End-of-life Min	End-of-life Max	Unit
DD	Power supply transient amplitude	XXI	Q	0	600	0	600	0	600	0	600	mV
EE	Power supply filter capacitance	XXII	S	1 6	7.2	I.6	7 2	I.6	7.2	I.6	7 2	µF
FF1-FF24	Input capacitance (data and disable)	XXII	Q	2	15	2	15	2	15	2	15	pF
FF25	Input capacitance (mode control)	XXII	Q	6	35	6	35	6	35	6	35	pF
GG1-GG12	Output capacitance	XXII	Q	4	30	4	30	4	30	4	30	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only

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4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

*4.3 Test equipment. The following test equipment is necessary to conduct acceptance tests for this module. Test equipment shall be as specified in MIL-STD-1665 and that listed below

Equipment	MIL-STD-1665 designator	No. required (maximum per test figure)
DC power supply	Item P01	4
Pulse generator	Item J01	1
Megohm bridge	Item R06	1
DC voltmeter	Item V03	1
DC ammeter	Item X01	1
Oscilloscope	Item D05	1
Oscilloscope probe	Item Y01	2
Capacitance bridge	Item C01	1
Capacitance meter	Item C05	1
Resistors		
127Ω ±1%, 1/4 W		12
51.1Ω ±1%, 1/4 W		1
75Ω ±1%, 1/4 W		12
55Ω ±1%, 3 W		1
54.9Ω ±1%, 1/2 W		1
127Ω ±1%, 1/2 W		1
1.13 kΩ ±1%, 1/4 W		1
50Ω ±5%, 1/2 W		1
51Ω ±5%, 1/2 W		1
10Ω ±5%, 1/2 W		1
1.21 kΩ ±1%, 1/8 W		12
Value as required for figure 14, note 3, 1/8 W		12
45Ω ±1%, 1/4 W		12
Value as required for figure 14, note 4, 1/8 W		12
150Ω ±1%, 1/4 W		6
Capacitors		
Value as required for figure 14, note 5		12
Value as required for figure 14, note 6		12
Value as required for figure 14, note 7		12
Value as required for figure 28, note 3		12
Diodes		
1N647, or equivalent		1
1N4150, or equivalent		12
Switch		
SPDT		1

All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table II.

* 4.4.1 Alternate test methods Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity (SEMP-QAA) and they conform to the requirements of MIL-M-28787.

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4.4.2 Order of testing. Test conditions A through GG shall be performed in any order with the following exceptions. If conditions A, B, C, D, or E are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, E where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through GG in table II shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions E through EE shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Transient thermal test. The transient thermal test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table II which are marked with an "X" in the test type column shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test, electrical tests E through GG shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirement of table II.

5. PACKAGING

* 5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table II, the module is considered to have completed its useful life (see 3.8).

* 6.3 Cross reference of classification. Part number M28787/47-1 supersedes part number M28787/47. The two parts are interchangeable.

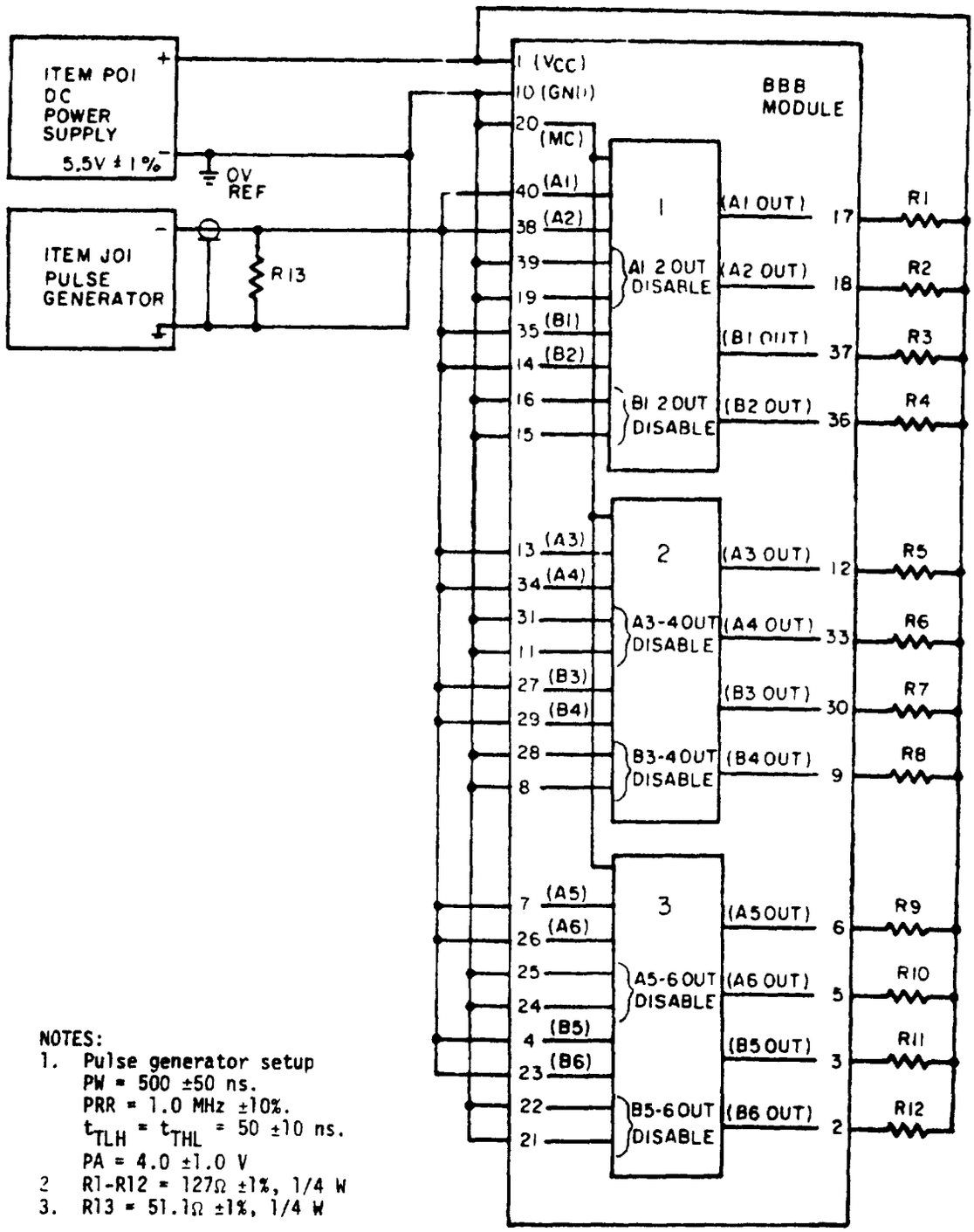
* 6.4 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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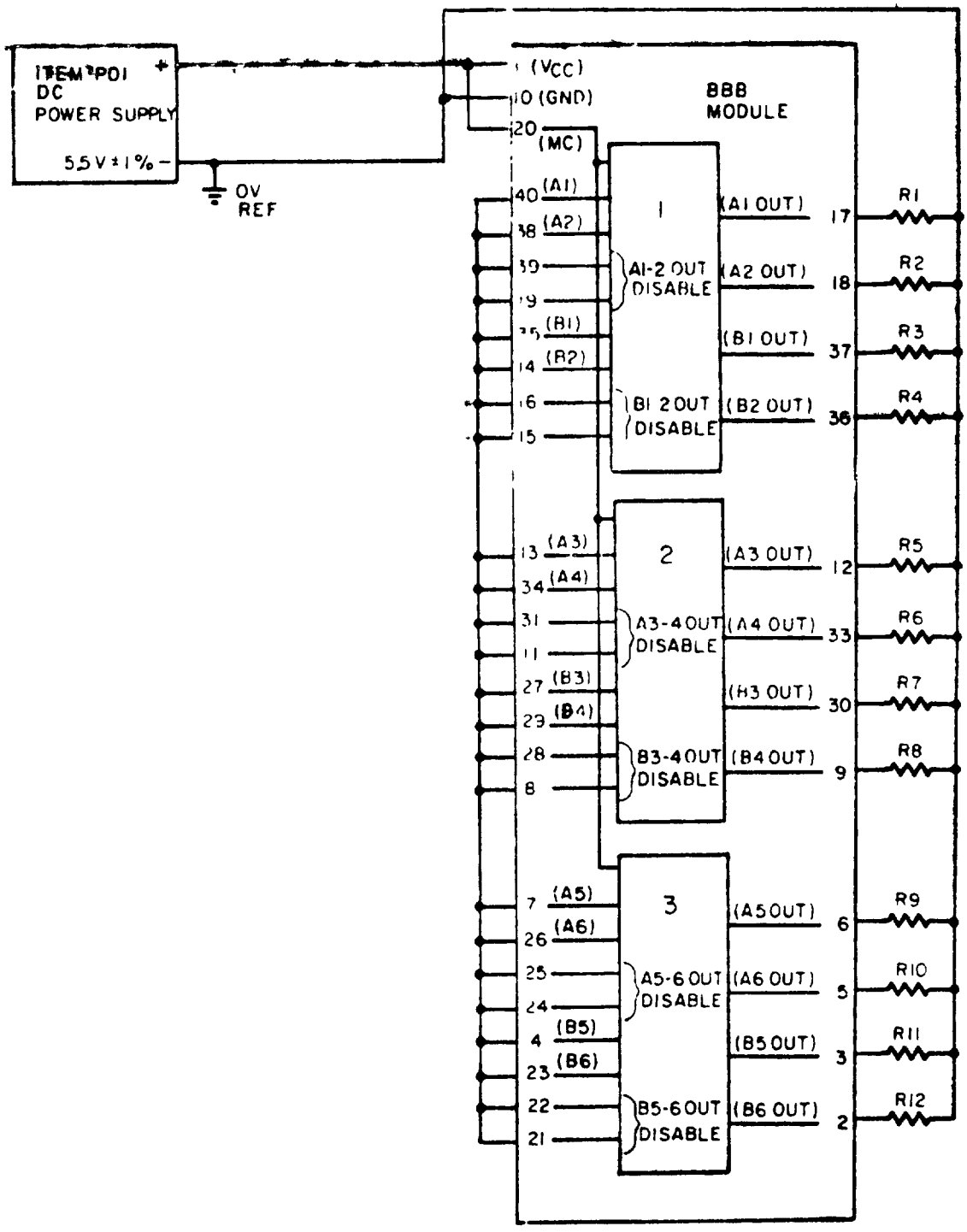
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NOTES:
1. Pulse generator setup
   PW = 500 ±50 ns.
   PRR = 1.0 MHz ±10%.
   $t_{TLH} = t_{THL}$ = 50 ±10 ns.
   PA = 4.0 ±1.0 V
2. R1-R12 = 127Ω ±1%, 1/4 W
3. R13 = 51.1Ω ±1%, 1/4 W

FIGURE 2 Operating temperature and life test circuit.

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NOTE R1-R12 = 75Ω ±1%, 1/4 W.

* FIGURE 3. Transient thermal test circuit.

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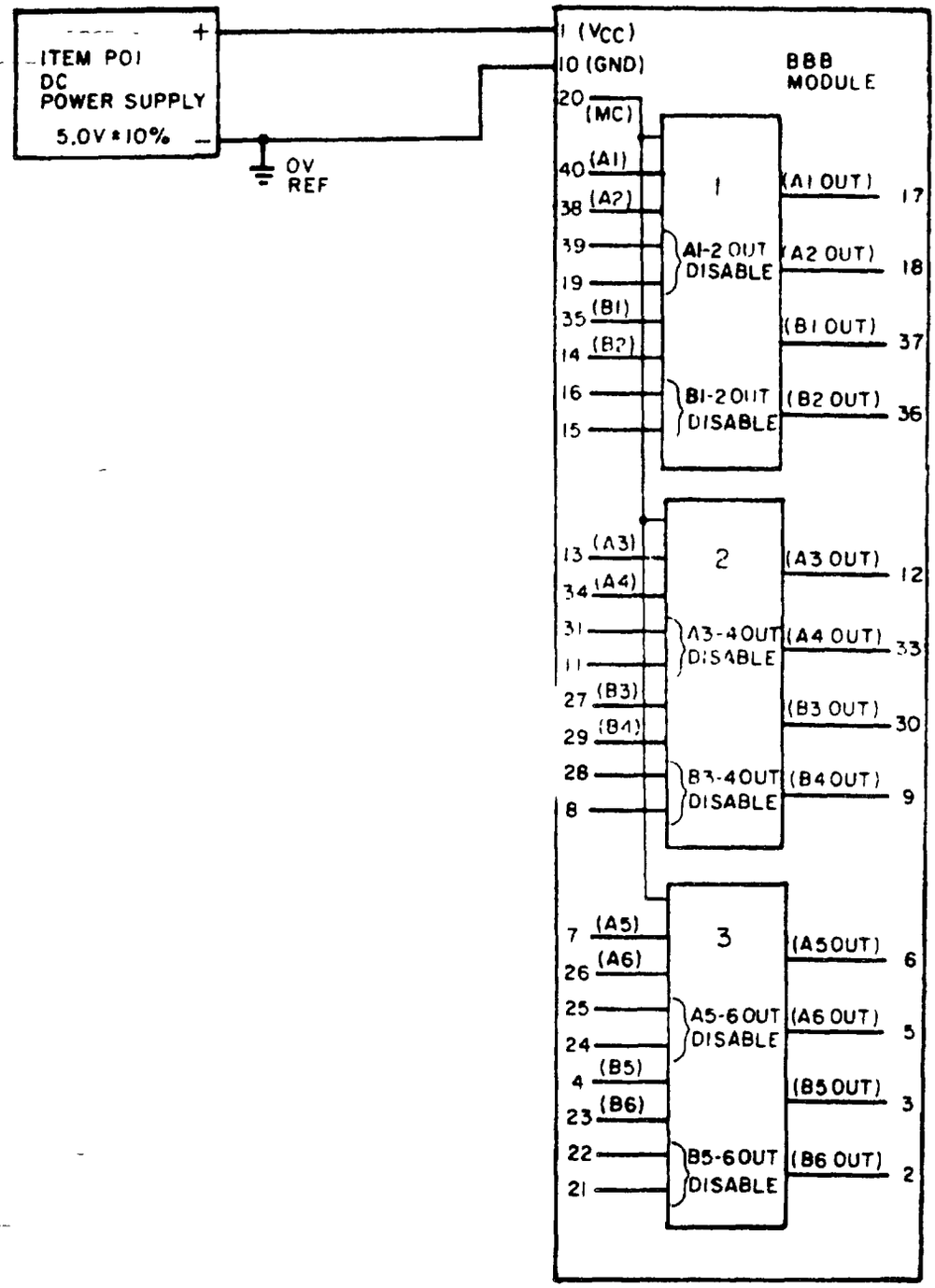


FIGURE 4. Vibration test circuit.

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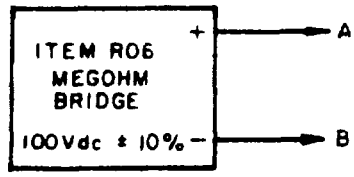
TABLE III. Isolation test.

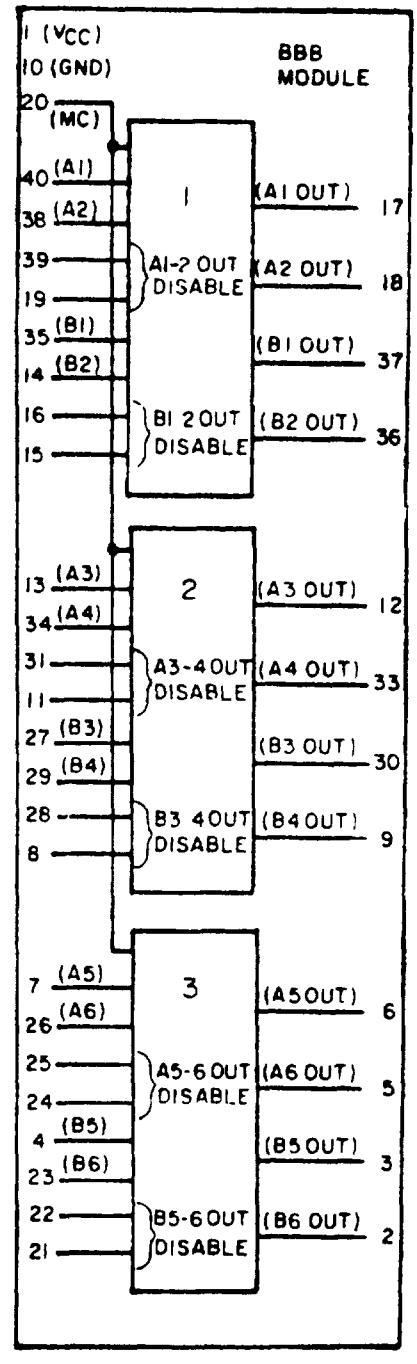
Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation, frame to module pins	All pins	Frame	10		MΩ	5
A2	Isolation, used to unused pins	All pins except pin 32	32	10		MΩ	5

TABLE IV. Maximum stress current check test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
B1	A1 OUT Maximum stress current check	17	See note 6 of figure 6			6
B2	A2 OUT	18				
B3	B1 OUT	37				
B4	B2 OUT	36				
B5	A3 OUT	12				
B6	A4 OUT	33				
B7	B3 OUT	30				
B8	B4 OUT	9				
B9	A5 OUT	6				
B10	A6 OUT	5				
B11	B5 OUT	3				
B12	B6 OUT	2				

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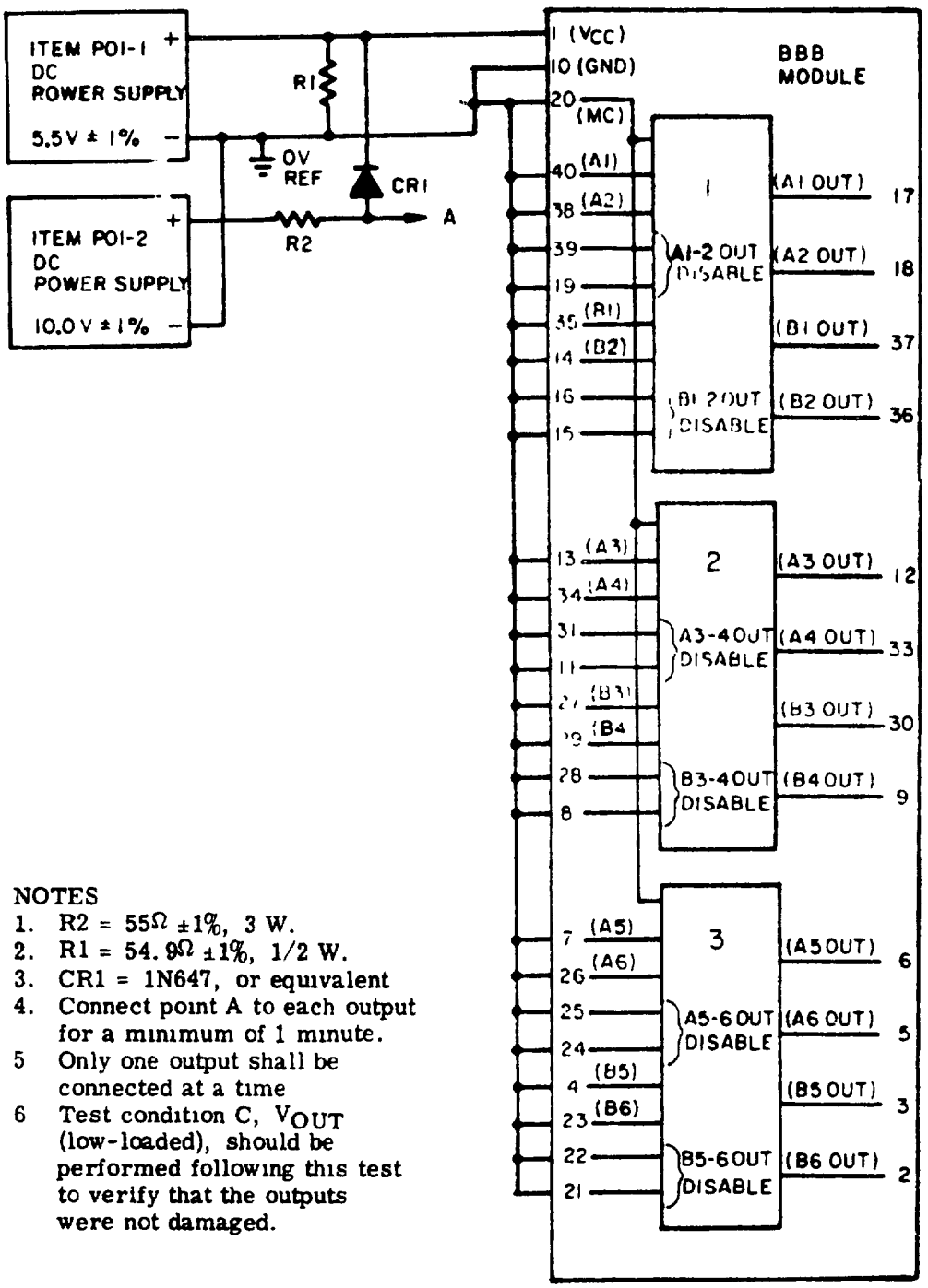
WARNING

When using the megohm bridge, high voltage may be present at any of the output binding posts. Always set the function switch of the megohm bridge to "discharge" before connecting or disconnecting the test leads.

NOTE. For test condition A1, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

* FIGURE 5. Isolation test circuit for table III.

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NOTES
1. R2 = 55Ω ±1%, 3 W.
2. R1 = 54.9Ω ±1%, 1/2 W.
3. CR1 = 1N647, or equivalent
4. Connect point A to each output for a minimum of 1 minute.
5 Only one output shall be connected at a time
6 Test condition C, V_{OUT} (low-loaded), should be performed following this test to verify that the outputs were not damaged.

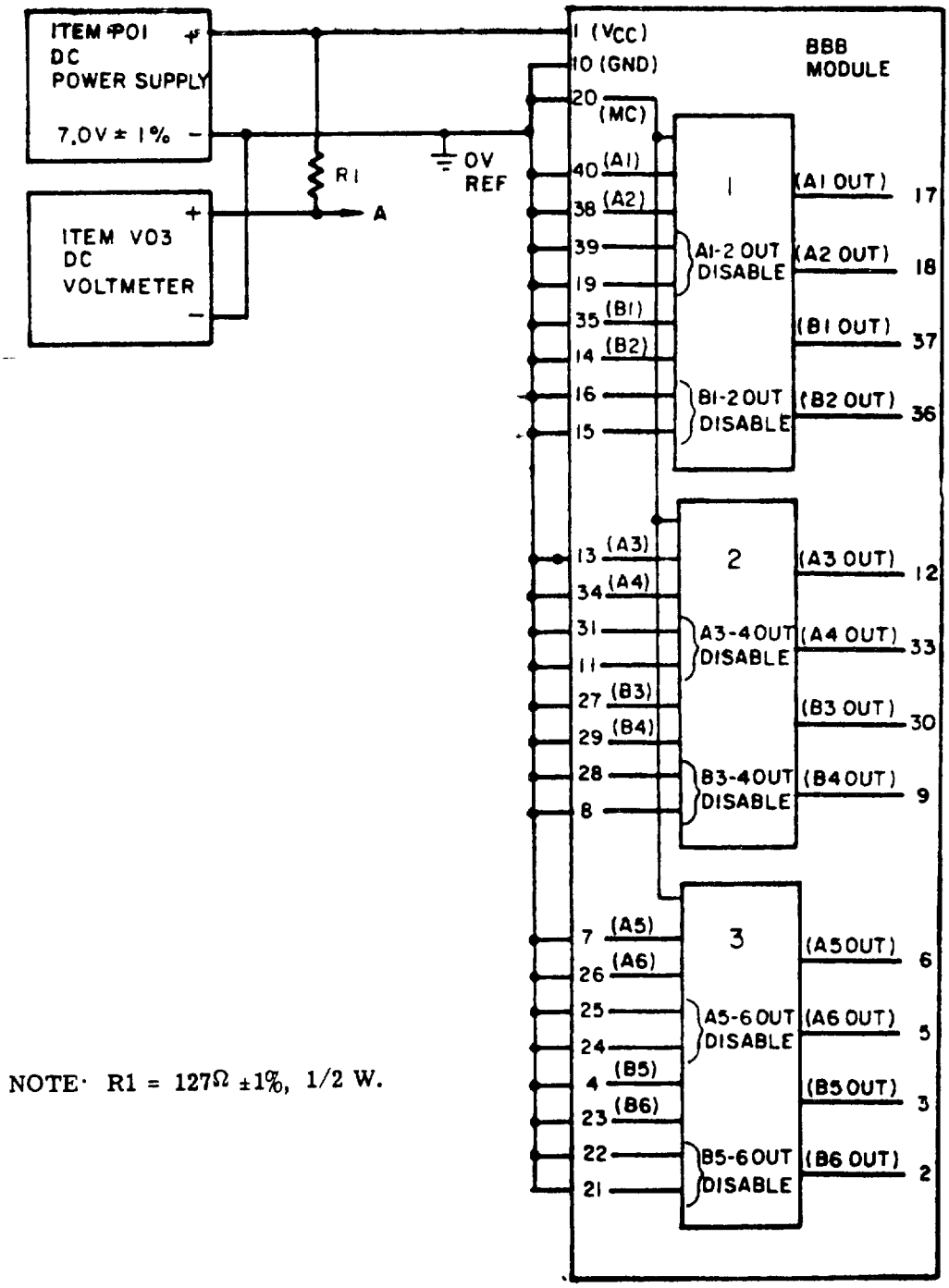
FIGURE 6. Maximum stress current check test circuit for table IV.

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TABLE V. V_{OUT} (low-loaded) test.

Test condition	Parameter		Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	Min	Max		
C1	A1 OUT	V_{OUT} (low-loaded)	17	0.01	1.0	V	7
C2	A2 OUT		18				
C3	B1 OUT		37				
C4	B2 OUT		36				
C5	A3 OUT		12				
C6	A4 OUT		33				
C7	B3 OUT		30				
C8	B4 OUT		9				
C9	A5 OUT		6				
C10	A6 OUT		5				
C11	B5 OUT		3				
C12	B6 OUT		2				

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NOTE: R1 = 127Ω ±1%, 1/2 W.

FIGURE 7. V_{OUT} (low-loaded) test circuit for table V.

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TABLE VI. Input breakdown current test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D1	A1 Input breakdown current	4,7,8,11,13,14,15,16, 19,20,21,22,23,24,25, 26,27,28,29,31,34,35, 38,39	40	0.0	1.0	mA	8
D2	A2	4,7,8,11,13,14,15,16, 19,20,21,22,23,24,25, 26,27,28,29,31,34,35, 39,40	38				
D3	B1	4,7,8,11,13,14,15,16, 19,20,21,22,23,24,25, 26,27,28,29,31,34,38, 39,40	35				
D4	B2	4,7,8,11,13,15,16,19, 20,21,22,23,24,25,26, 27,28,29,31,34,35,38, 39,40	14				
D5	A1-2 OUT DISABLE	4,7,8,11,13,14,15,16, 19,20,21,22,23,24,25, 26,27,28,29,31,34,35, 38,40	39				
D6	A1-2 OUT DISABLE	4,7,8,11,13,14,15,16, 20,21,22,23,24,25,26, 27,28,29,31,34,35,38, 39,40	19				
D7	B1-2 OUT DISABLE	4,7,8,11,13,14,15,19, 20,21,22,23,24,25,26, 27,28,29,31,34,35,38, 39,40	16				
D8	B1-2 OUT DISABLE	4 7,8,11,13,14,16,19, 20,21,22,23,24,25,26, 27,28,29,31,34,35,38, 39,40	15				

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TABLE VI. Input breakdown current test - Continued.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Initial limits 25° ±5°C Ambient: Min	Initial limits 25° ±5°C Ambient: Max	Unit	Test figure
D9	A3 Input breakdown current	4,7,8,11,14,15,16,19, 20,21,22,23,24,25,26, 27,28,29,31,34,35,38, 39,40	13	0.0	1.0	mA	8
D10	A4	4,7,8,11,13,14,15,16, 19,20,21,22,23,24,25, 26,27,28,29,31,35,38, 39,40	34				
D11	B3	4,7,8,11,13,14,15,16, 19,20,21,22,23,24,25, 26,28,29,31,34,35,38, 39,40	27				
D12	B4	4,7,8,11,13,14,15,16, 19,20,21,22,23,24,25, 26,27,28,31,34,35,38, 39,40	29				
D13	A3-4 OUT DISABLE	4,7,8,11,13,14,15,16, 19,20,21,22,23,24,25, 26,27,28,29,34,35,38, 39,40	31				
D14	A3-4 OUT DISABLE	4,7,8,13,14,15,16,19, 20,21,22,23,24,25,26, 27,28,29,31,34,35,38, 39,40	11				
D15	B3-4 OUT DISABLE	4,7,8,11,13,14,15,16, 19,20,21,22,23,24,25, 26,27,29,31,34,35,38, 39,40	28				

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TABLE VI. Input breakdown current test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D16	B3-4 OUT Input breakdown current DISABLE	4,7,11,13,14,15,16,19, 20,21,22,23,24,25,26, 27,28,29,31,34,35,38, 39,40	8	0.0	1.0	mA	8
D17	A5	4,8,11,13,14,15,16,19, 20,21,22,23,24,25,26, 27,28,29,31,34,35,38, 39,40	7				
D18	A6	4,7,8,11,13,14,15,16, 19,20,21,22,23,24,25, 27,28,29,31,34,35,38, 39,40	26				
D19	B5	7,8,11,13,14,15,16,19, 20,21,22,23,24,25,26, 27,28,29,31,34,35,38, 39,40	4				
D20	B6	4,7,8,11,13,14,15,16, 19,20,21,22,24,25,26, 27,28,29,31,34,35,38, 39,40	23				
D21	A5-6 OUT DISABLE	4,7,8,11,13,14,15,16, 19,20,21,22,23,24,26, 27,28,29,31,34,35,38, 39,40	25				
D22	A5-6 OUT DISABLE	4,7,8,11,13,14,15,16, 19,20,21,22,23,25,26, 27,28,29,31,34,35,38, 39,40	24				

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TABLE VI. Input breakdown current test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D23	B5-6 OUT Input breakdown current DISABLE	4,7,8,11,13,14,15,16, 19,20,21,23,24,25,26, 27,28,29,31,34,35,38, 39,40	22	0.0	1.0	mA	8
D24	B5-6 OUT Input breakdown current DISABLE	4,7,8,11,13,14,15,16, 19,20,22,23,24,25,26, 27,28,29,31,34,35,38, 39,40	21	0.0	1.0	mA	8
D25	MC Input breakdown current	4,7,8,11,13,14,15,16, 19,21,22,23,24,25,26, 27,28,29,31,34,35,38, 39,40	20	0.0	3.0	mA	8

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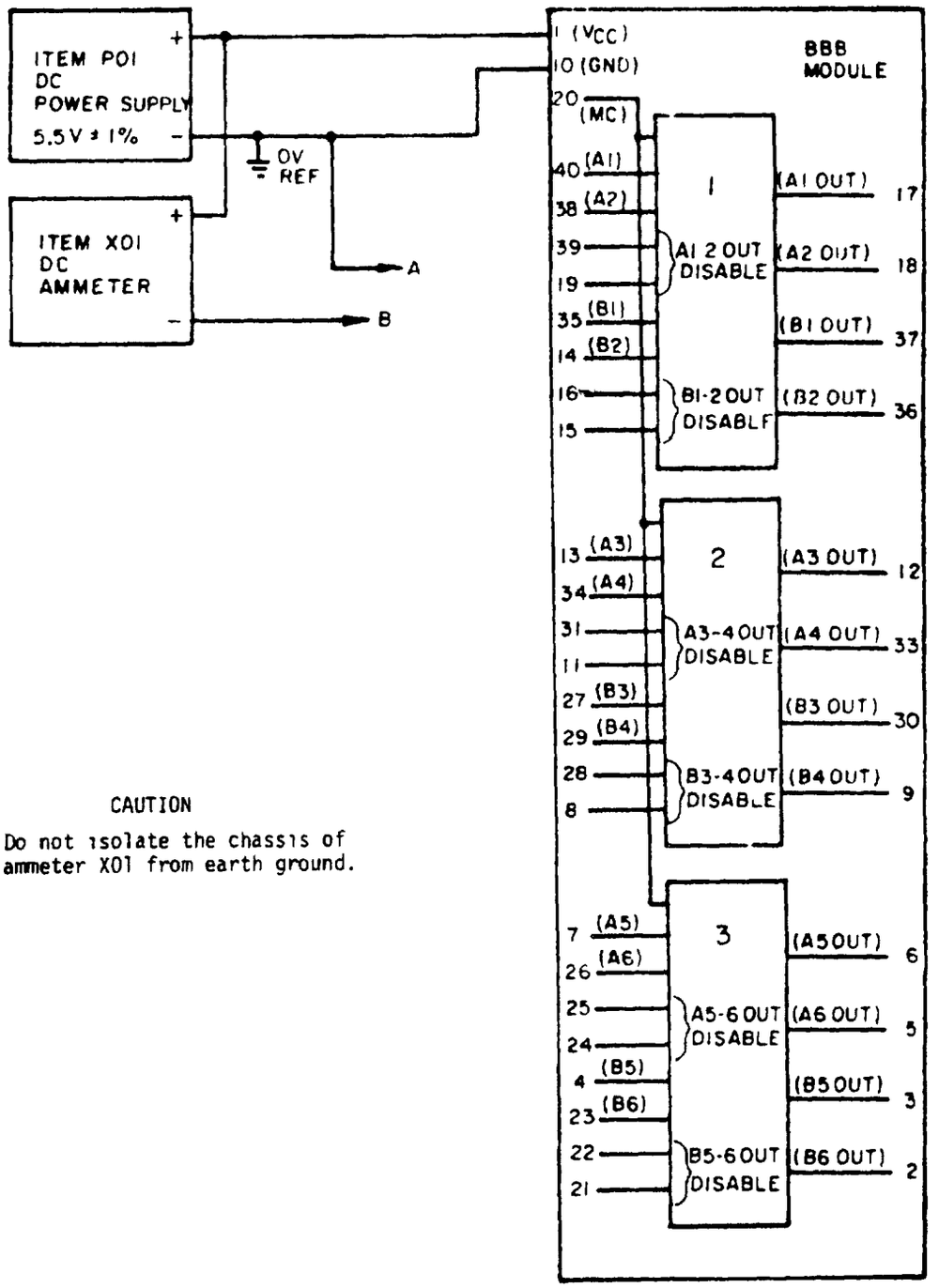


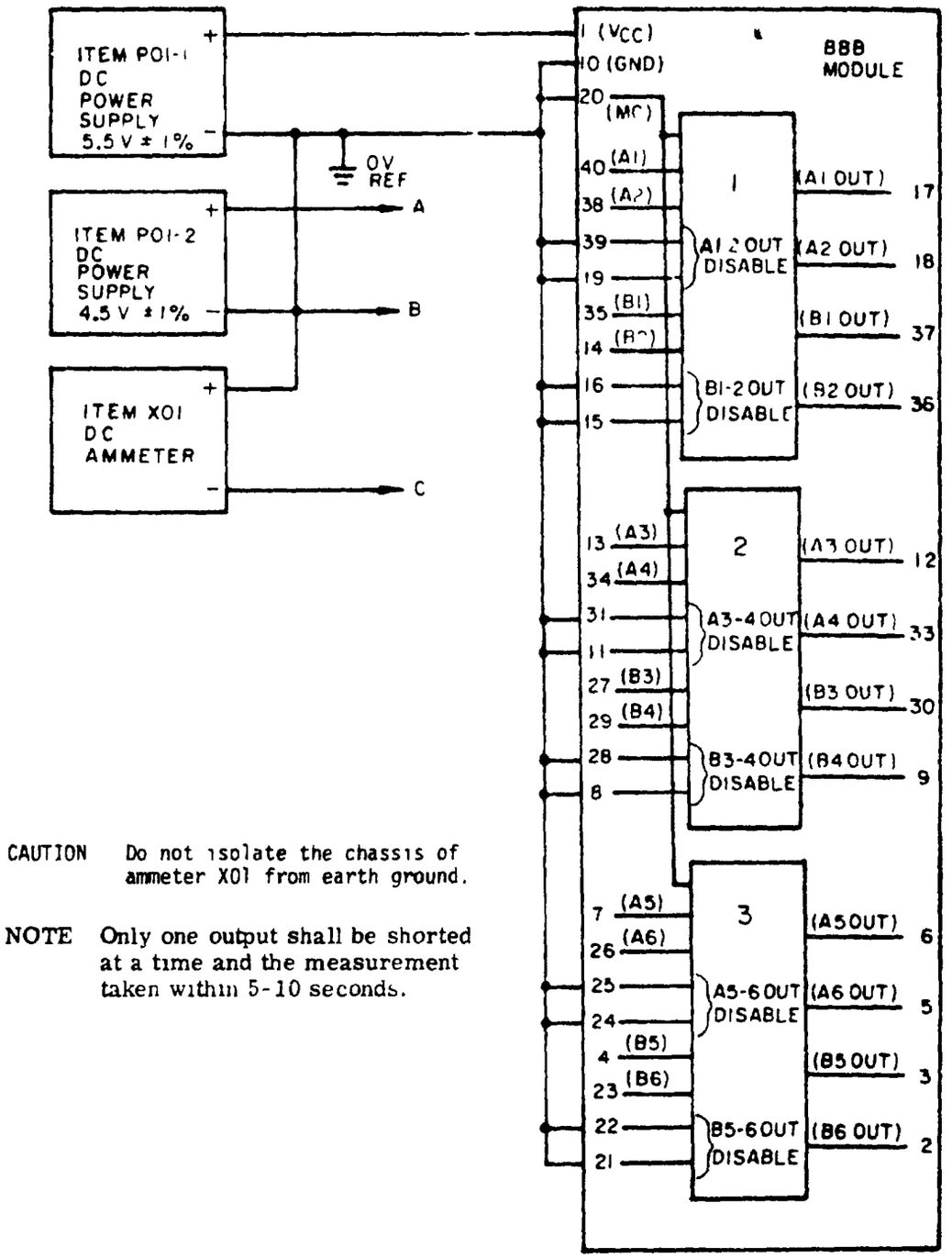
FIGURE 8. Input breakdown current test circuit for table VI.

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TABLE VII. Output short circuit current test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
E1	A1 OUT Output short circuit current	40	4,7,13,14,23,26,27, 29,34,35,38	17	-40	-120	mA	9
E2	A2 OUT	38	4,7,13,14,23,26,27, 29,34,35,40	18				
E3	B1 OUT	35	4,7,13,14,23,26,27, 29,34,38,40	37				
E4	B2 OUT	14	4,7,13,23,26,27,29, 34,35,38,40	36				
E5	A3 OUT	13	4,7,14,23,26,27,29, 34,35,38,40	12				
E6	A4 OUT	34	4,7,13,14,23,26,27, 29,35,38,40	33				
E7	B3 OUT	27	4,7,13,14,23,26,29, 34,35,38,40	30				
E8	B4 OUT	29	4,7,13,14,23,26,27, 34,35,38,40	9				
E9	A5 OUT	7	4,13,14,23,26,27,29, 34,35,38,40	6				
E10	A6 OUT	26	4,7,13,14,23,27,29, 34,35,38,40	5				
E11	B5 OUT	4	7,13,14,23,26,27,29, 34,35,38,40	3				
E12	B6 OUT	23	4,7,13,14,26,27,29, 34,35,38,40	2				

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CAUTION Do not isolate the chassis of ammeter X01 from earth ground.

NOTE Only one output shall be shorted at a time and the measurement taken within 5-10 seconds.

FIGURE 9 Output short circuit current test circuit for table VII.

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TABLE VIII Input and output clamp diode voltage test.

Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
F1	A1	Input clamp diode voltage	1	40	-0.4	-1.5	V	10
F2	A2			38				
F3	B1			35				
F4	B2			14				
F5	A1-2 OUT DISABLE			39				
F6	A1-2 OUT DISABLE			19				
F7	B1-2 OUT DISABLE			16				
F8	B1-2 OUT DISABLE			15				
F9	A3			13				
F10	A4			34				
F11	B3			27				
F12	B4			29				
F13	A3-4 OUT DISABLE			31				
F14	A3-4 OUT DISABLE			11				
F15	B3-4 OUT DISABLE			28				

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TABLE VIII. Input and output clamp diode voltage test - Continued.

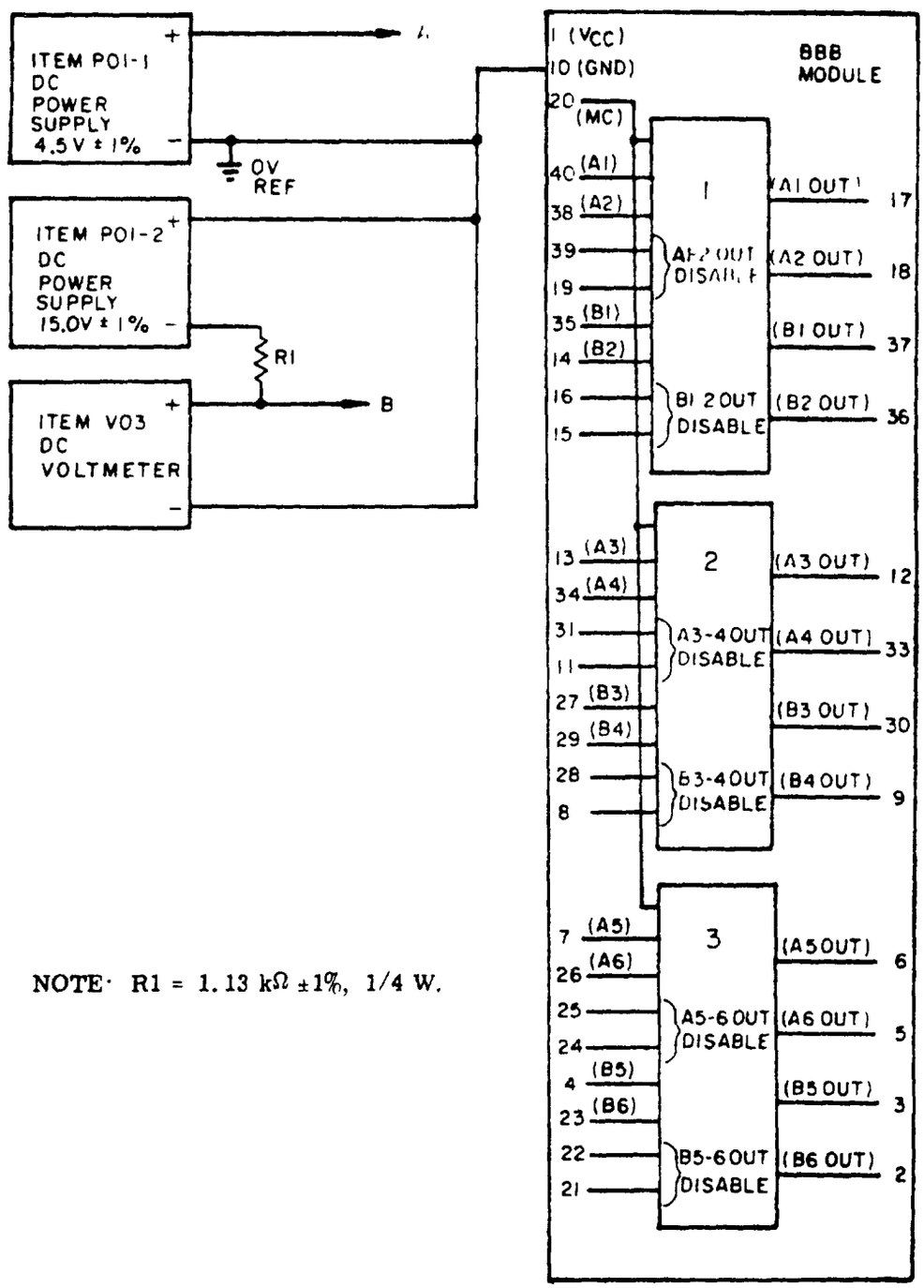
Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
F16	B3-4 OUT DISABLE	Input clamp diode voltage	1	8	-0.4	-1.5	V	10
F17	A5			7				
F18	A6			26				
F19	B5			4				
F20	B6			23				
F21	A5-6 OUT DISABLE			25				
F22	A5-6 OUT DISABLE			24				
F23	B5-6 OUT DISABLE			22				
F24	B5-6 OUT DISABLE			21				
F25	MC			20				
G1	A1 OUT	Output clamp diode voltage	Open	17				
G2	A2 OUT			18				
G3	B1 OUT			37				
G4	B2 OUT			36				
G5	A3 OUT			12				

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TABLE VIII. Input and output clamp diode voltage test - Continued.

Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
G6	A4 OUT	Output clamp diode voltage	Open	33	-0.4	-1.5	V	10
G7	B3 OUT			30				
G8	B4 OUT			9				
G9	A5 OUT			6				
G10	A6 OUT			5				
G11	B5 OUT			3				
G12	B6 OUT			2				

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NOTE: R1 = 1.13 kΩ ±1%, 1/4 W.

FIGURE 10. Input and output clamp diode voltage test circuit for table VIII.

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TABLE IX Output leakage current (high-level and power-off) test

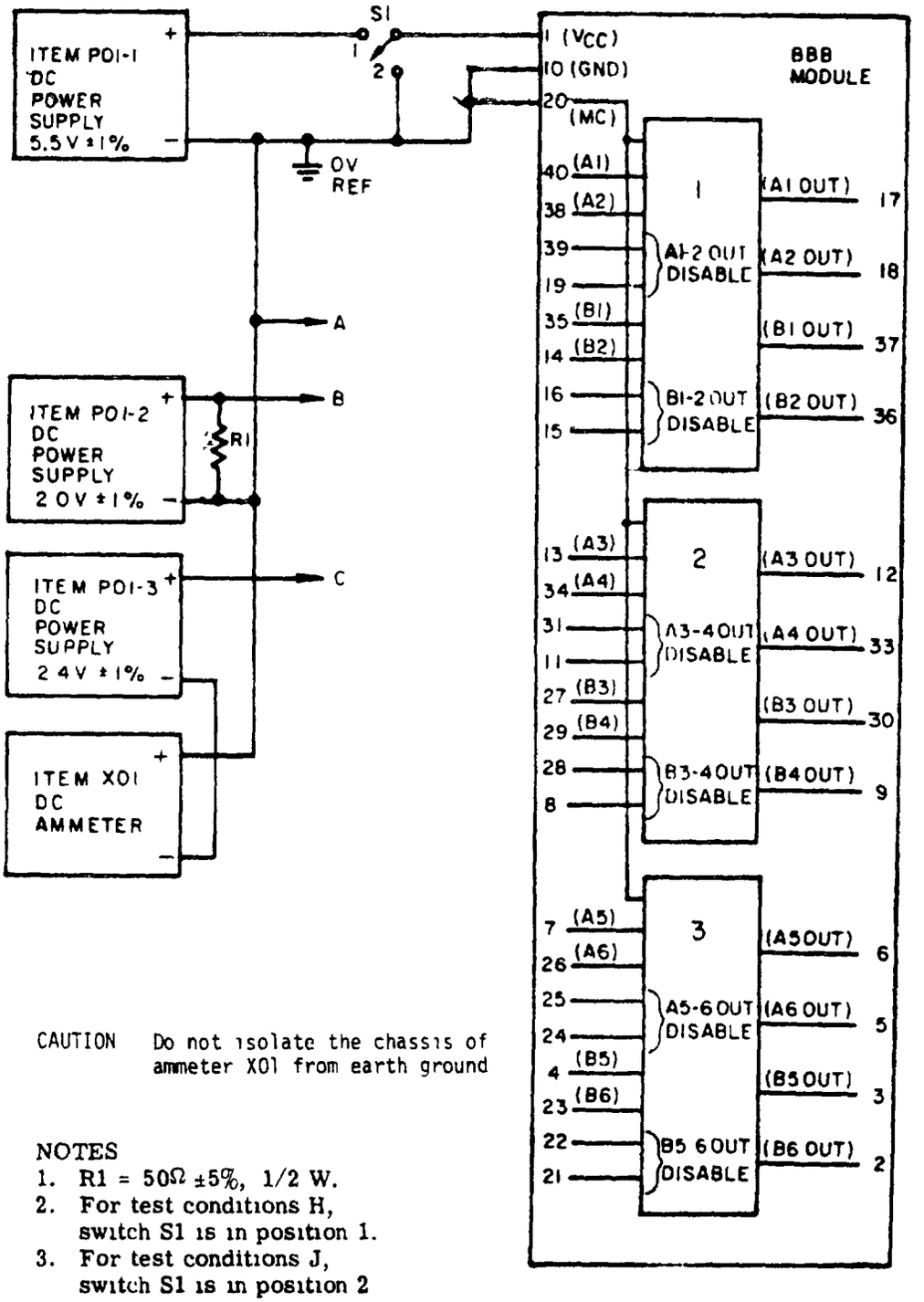
Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
H1	A1 OUT	Output leakage current, high level	19,40	39	17	-40	40	µA	11
H2	A2 OUT		19,38	39	18				
H3	B1 OUT		15,35	16	37				
H4	B2 OUT		14,15	16	36				
H5	A3 OUT		11,13	31	12				
H6	A4 OUT		11,34	31	33				
H7	B3 OUT		8,27	28	30				
H8	B4 OUT		8,29	28	9				
H9	A5 OUT		7,24	25	6				
H10	A6 OUT		24,26	25	5				
H11	B5 OUT		4,21	22	3				
H12	B6 OUT		21,23	22	2				
J1	A1 OUT	Output leakage current, power off	Open	Open	17	0			
J2	A2 OUT				18				
J3	B1 OUT				37				
J4	B2 OUT				36				
J5	A3 OUT				12				

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TABLE IX Output leakage current (high-level and power-off) test - Continued

Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
J6	A4 OUT	Output leakage current, power off	Open	Open	33	0	40	µA	11
J7	B3 OUT	↓	↓	↓	30	↓	↓	↓	↓
J8	B4 OUT				9				
J9	A5 OUT				6				
J10	A6 OUT				5				
J11	B5 OUT				3				
J12	B6 OUT				2				

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CAUTION Do not isolate the chassis of ammeter X01 from earth ground

NOTES
1. R1 = 50Ω ±5%, 1/2 W.
2. For test conditions H, switch S1 is in position 1.
3. For test conditions J, switch S1 is in position 2

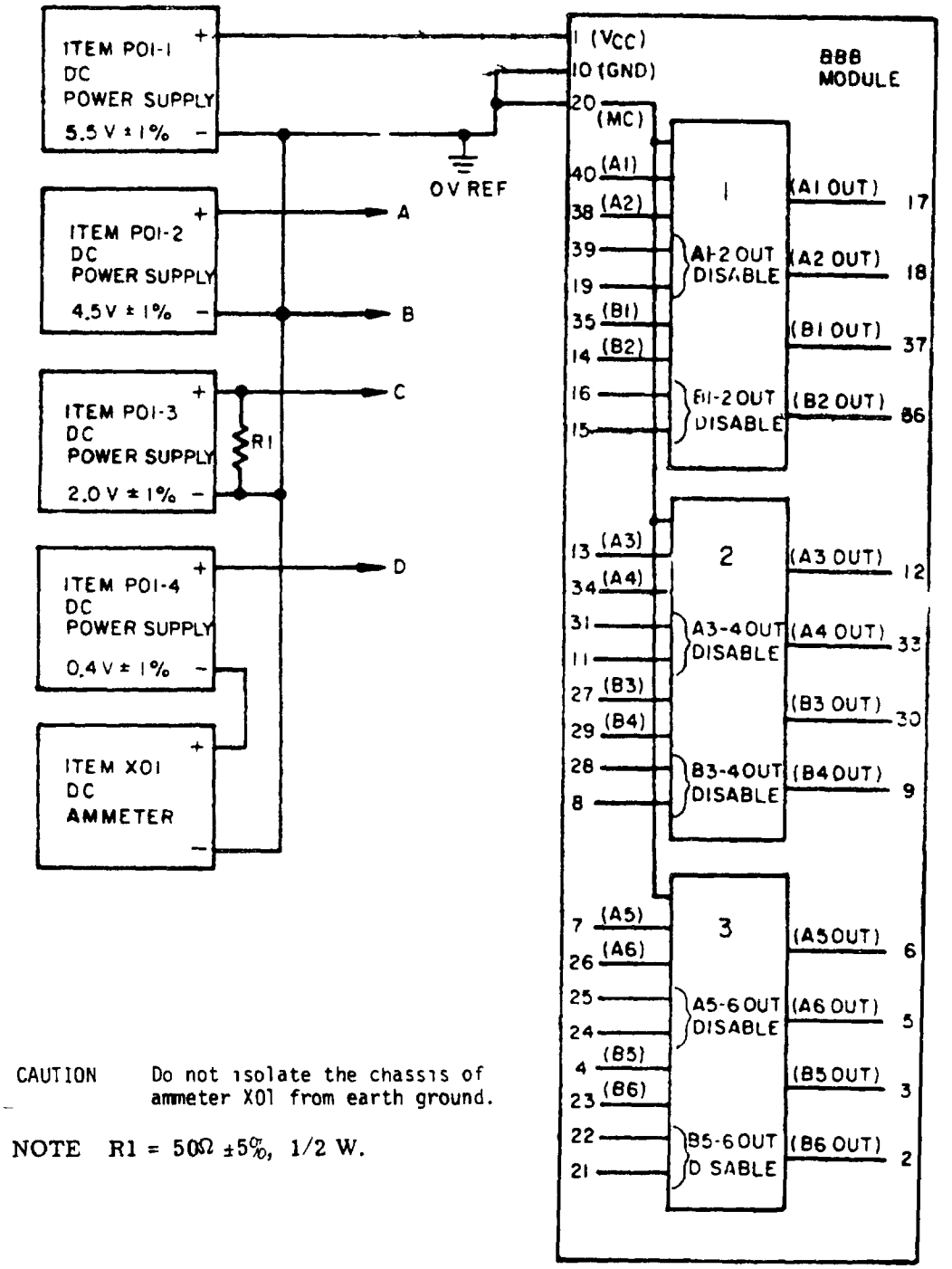
FIGURE 11 Output leakage current (high-level and power-off) test circuit for table IX.

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TABLE X. Output leakage current (low-level) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
K1	A1 OUT Output leakage current, low level	40	39	19	17	-40	40	µA	12
K2	A2 OUT	38	39	19	18				
K3	B1 OUT	35	16	15	37				
K4	B2 OUT	14	16	15	36				
K5	A3 OUT	13	31	11	12				
K6	[illegible] OUT	34	31	11	33				
K7	B3 OUT	27	28	8	30				
K8	B4 OUT	29	28	8	9				
K9	A5 OUT	7	25	24	6				
K10	A6 OUT	26	25	24	5				
K11	B5 OUT	4	22	21	3				
K12	B6 OUT	23	22	21	2				

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CAUTION Do not isolate the chassis of ammeter XO1 from earth ground.

NOTE R1 = 50Ω ±5%, 1/2 W.

FIGURE 12. Output leakage current (low-level) test circuit for table X.

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TABLE XI Output voltage (low-level and high-level) test.

Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
L1	A1 OUT	Output voltage, low level (single-ended mode) I_{OL} = 32 mA	14,35,38	20,40	17	0.0	0.4	V	13
L2	A2 OUT		14,35,40	20,38	18				
L3	B1 OUT		14,38,40	20,35	37				
L4	B2 OUT		35,38,40	14,20	36				
L5	A3 OUT		27,29,34	13,20	12				
L6	A4 OUT		13,27,29	20,34	33				
L7	B3 OUT		13,29,34	20,27	30				
L8	B4 OUT		13,27,34	20,29	9				
L9	A5 OUT		4,23,26	7,20	6				
L10	A6 OUT		4,7,23	20,26	5				
L11	B5 OUT		7,23,26	4,20	3				
L12	B6 OUT		4,7,26	20,23	2				
L13	A1 OUT	Output voltage, low level (differential mode) I_{OL} = 32 mA	20,35,40	14,38	17				
L14	A2 OUT				18				
L15	B1 OUT				37				
L16	B2 OUT				36				

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TABLE XI Output voltage (low-level and high-level) test - Continued

Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
L17	A3 OUT	Output voltage, low level (differential mode) I_{OL} = 32 mA	20,13,27	29,34	12	0.0	0.4	V	13
L18	A4 OUT				33				
L19	B3 OUT				30				
L20	B4 OUT				9				
L21	A5 OUT		4,7,20	23,26	6				
L22	A6 OUT				5				
L23	B5 OUT				3				
L24	B6 OUT				2				
L25	A1 OUT	Output voltage, low level (I_{OL} = 40 mA)	14,35,38	20,40	17		0.5		
L26	A2 OUT		14,35,40	20,38	18				
L27	B1 OUT		14,38,40	20,35	37				
L28	B2 OUT		35,38,40	14,20	36				
L29	A3 OUT		27,29,34	13,20	12				
L30	A4 OUT		13,27,29	20,34	33				
L31	B3 OUT		13,29,34	20,27	30				
L32	B4 OUT		13,27,34	20,29	9				

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TABLE XI Output voltage (low-level and high-level) test - Continued.

Test condition	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25° ±5°C Ambient Min	Initial limits 25° ±5°C Ambient Max	Unit	Test figure
L33	A5 OUT	Output voltage, low level (I_{OL} = 40 mA)	4,23,26	7,20	6	0.0	0 5	V	13
L34	A6 OUT		4,7,23	20,26	5				
L35	B5 OUT		7,23,26	4,20	3				
L36	B6 OUT		4,7,26	20,23	2				
M1	A1 OUT	Output voltage, high level (single-ended mode) I_{OH} = -2 mA)	40	4,7,13,14, 20,23,26, 27,29,34, 35,38	17	2 4	4 5		
M2	A2 OUT		40	4,7,13,14, 20,23,26, 27 29,34, 35,40	18				
M3	B1 OUT		35	4,7,13,14, 20,23,26, 27,29,34, 38,40	37				
M4	B2 OUT		14	4,7,13,20, 23,26,27, 29,34,35, 38,40	36				
M5	A3 OUT		13	4,7,14,20, 23,26,27, 29,34,35, 38,40	12				
M6	A4 OUT		34	4,7,13,14, 20,23,26, 27,29,35 38,40	33				

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TABLE XI Output voltage (low-level and high-level) test - Continued

Test condition		Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
M7	B3 OUT	Output voltage, high level (single-ended mode) I_{OH} = -2 mA	27	4,7,13,14, 20,23,26, 29,34,35, 38,40	30	2 4	4 5	V	13
M8	B4 OUT		29	4,7,13,14, 20,23,26, 27,34,35, 38,40	9				
M9	A5 OUT		7	4,13,14, 20,23,26, 27,29,34, 35,38,40	6				
M10	A6 OUT		26	4,7,13,14, 20,23,27, 29,34,35, 38,40	5				
M11	B5 OUT		4	7,13,14, 20,23,26, 27,29,34, 35,38,40	3				
M12	B6 OUT		23	4,7,13,14, 20,26,27, 29,34,35, 38,40	2				
M13	A1 OUT	Output voltage, high level (differential mode) I_{OH} = -2 mA	4,7,13,20, 27,35	14,23,26, 29,34,38, 40	17				
M14	A2 OUT	Output voltage, high level (differential mode) I_{OH} = -2 mA	4,7,13,20, 27,35,38, 40	14,23,26, 29,34	18				

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TABLE XI Output voltage (low-level and high-level) test - Continued

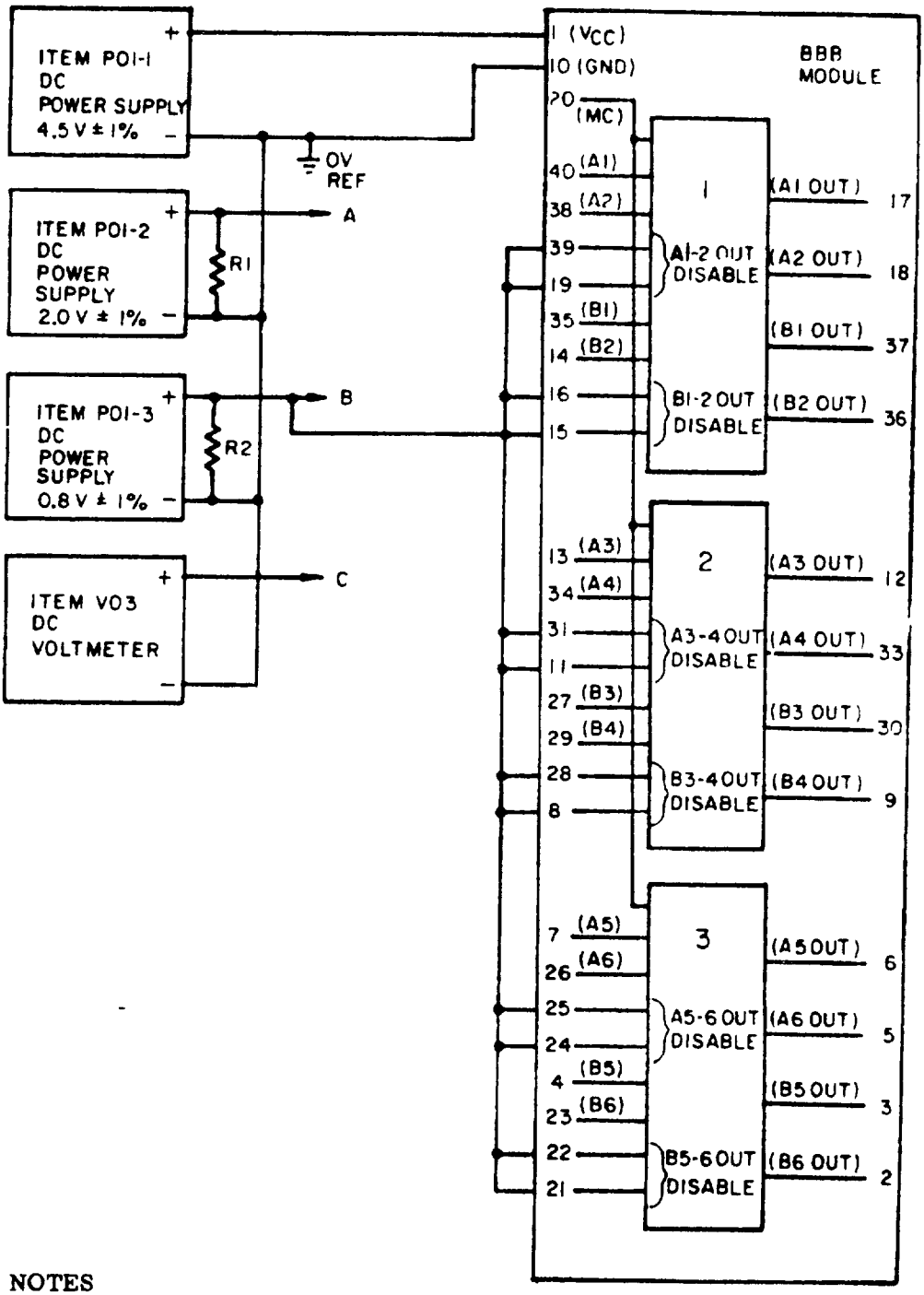
Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
M15	B1 OUT	Output voltage, high level (differential mode) I_{OH} = -2 mA	4,7,13,20, 27,40	14,23,26, 29,34,35, 38	37	2 4	4 5	V	13
M16	B2 OUT		4,7,13,14, 20,27,35, 40	23,26,29, 34,38	36				
M17	A3 OUT		4,7,20,27, 35,40	13,14,23, 26,29,34, 38	12				
M18	A4 OUT		4,7,13,20, 27,34,35, 40	14,23,26, 29,38	33				
M19	B3 OUT		4,7,13,20, 35,40	14,23,26, 27,29,34, 38	30				
M20	B4 OUT		4,7,13,20, 27,29,35, 40	14,23,26, 34,38	9				
M21	A5 OUT		4,13,20, 27,35,40	7,14,23, 26,29,34, 38	6				
M22	A6 OUT		4,7,13,20, 26,27,35, 40	14,23,29, 34,38	5				
M23	B5 OUT		7,13,20, 27,35,40	4,14,23, 26,29,34, 38	3				

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TABLE XI Output voltage (low-level and high-level) test - Continued.

Test condition	Parameter		Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
M24	B6 OUT	Output voltage, high level (differential mode) I_{OH} = -2 mA	4,7,13,20, 23,27,35, 40	14,26,29, 34,38	2	2.4	4.5	V	13
M25	A1 OUT	Output voltage, high level(I_{OH} = -40 mA)	14,20,38	35,40	17	1.8			
M26	A2 OUT				18				
M27	B1 OUT				37				
M28	B2 OUT				36				
M29	A3 OUT		20,29,34	13,27	12				
M30	A4 OUT				33				
M31	B3 OUT				30				
M32	B4 OUT				9				
M33	A5 OUT		20,23,26	4,7	6				
M34	A6 OUT				5				
M35	B5 OUT				3				
M36	B6 OUT				2				

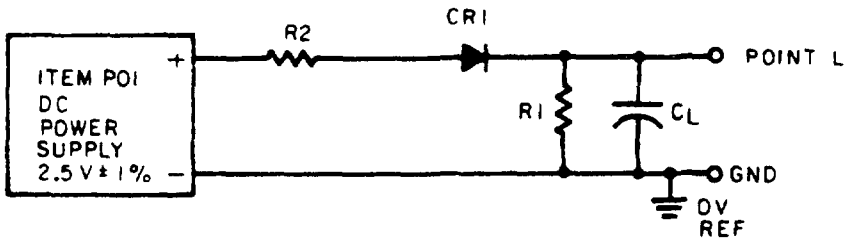
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NOTES
1. R1 = 51Ω ±5%, 1/2 W.
2. R2 = 10Ω ±5%, 1/2 W.
3. Load output under test with load as specified on figure 14.

FIGURE 13. Output voltage (low-level and high-level) test circuit for table XI.

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NOTES
1. CR1 = 1N4150, or equivalent.
2 R1 = 1.21 kΩ ±1%, 1/8 W.
3 R2 = Select value for a current flow of 32 0 mA ±5% out of point L with point L held at 0 4 V
4 For test conditions L25-L36 and M25-M36
R1 = 45Ω ±1%, 1/4 W.
R2 = Select value for a current flow of 40 mA ±5% out of point L with point L held at 0.5 V
5 For test conditions R, S, T1-T18, and U1-U18
C_L = 100 ±10 pF (including probe and parasitic capacitance).
6 For test conditions T19-T36 and U19-U36
C_L = 1,000 ±10 pF (including probe and parasitic capacitance)
7 For test conditions V, W, X, Y, Z, AA, and BB
C_L = 50 ±5 pF (including probe and paras tic capacitance)
8 Capacitor C_L is not required for dc electrical test conditions L and M
9 Lead length from load to module output pin under test is to be less than 1.00 inch (25 4 mm)

* FIGURE 14 BBB logic load circuit

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TABLE XII. Input current (high-level) test

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
N1	A1 Input current, high level	40	4,7,8,11,13,14,15,16,19,20, 21,22,23,24,25,26,27,28,29, 31,34,35,38,39	0	40	µA	15
N2	A2	38	4,7,8,11,13,14,15,16,19,20, 21,22,23,24,25,26,27,28,29, 31,34,35,39,40				
N3	B1	35	4,7,8,11,13,14,15,16,19,20, 21,22,23,24,25,26,27,28,29, 31,34,38,39,40				
N4	B2	14	4,7,8,11,13,15,16,19,20,21, 22,23,24,25,26,27,28,29,31, 34,35,38,39,40				
N5	A1-2 OUT DISABLE	39	4,7,8,11,13,14,15,16,19,20, 21,22,23,24,25,26,27,28,29, 31,34,35,38,40				
N6	A1-2 OUT DISABLE	19	4,7,8,11,13,14,15,16,20,21 22,23,24,25,26,27,28,29,31, 34,35,38,39,40				
N7	B1-2 OUT DISABLE	16	4,7,8,11,13,14,15,19,20,21, 22,23,24,25,26,27,28,29,31, 34,35,38,39,40				
N8	B1-2 OUT DISABLE	15	4,7,8,11,13,14,16,19,20,21, 22,23,24,25,26,27,28,29,31, 34,35,38,39,40				
N9	A3	13	4,7,8,11,14,15,16,19,20,21, 22,23,24,25,26,27,28,29,31, 34,35,38,39,40				

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TABLE XII Input current (high-level) test - Continued

Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
N10	A4	Input current, high level	34	4,7,8,11,13,14,15,16,19,20, 21,22,23,24,25,26,27,28,29, 31,35,38,39,40	0	40	μA	15
N11	B3		27	4,7,8,11,13,14,15,16,19,20, 21,22,23,24,25,26,28,29,31, 34,35,38,39,40				
N12	B4		29	4,7,8,11,13,14,15,16,19,20, 21,22,23,24,25,26,27,28,31, 34,35,38,39,40				
N13	A3-4 OUT DISABLE		31	4,7,8,11,13,14,15,16,19,20, 21,22,23,24,25,26,27,28,29, 34,35,38,39,40				
N14	A3-4 OUT DISABLE		11	4,7,8,13,14,15,16,19,20,21, 22,23,24,25,26,27,28,29,31, 34,35,38,39,40				
N15	B3-4 OUT DISABLE		28	4,7,8,11,13,14,15,16,19,20, 21,22,23,24,25,26,27,29,31, 34,35,38,39,40				
N16	B3-4 OUT DISABLE		8	4,7,11,13,14,15,16,19,20,21, 22,23,24,25,26,27,28,29,31, 34,35,38,39,40				
N17	A5		7	4,8,11,13,14,15,16,19,20,21, 22,23,24,25,26,27,28,29,31, 34,35,38,39,40				
N18	A6		26	4,7,8,11,13,14,15,16,19,20, 21,22,23,24,25,27,28,29,31, 34,35 38,39,40				

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TABLE XII Input current (high-level) test - Continued.

Test condition	Parameter		Connect A to pin	Connect B to pin	Initial limits 25° ±5°C Ambient Min	Initial limits 25° ±5°C Ambient Max	Unit	Test figure
N19	B5	Input current, high level	4	7,8,11,13,14,15,16,19,20,21, 22,23,24,25,26,27,28,29,31, 34,35,38,39,40	0	40	µA	15
N20	B6		23	4,7,8,11,13,14,15,16,19,20, 21,22,24,25,26,27,28,29,31, 34,35,38,39,40				
N21	A5-6 OUT DISABLE		25	4,7,8,11,13,14,15,16,19,20, 21,22,23,24,26,27,28,29,31, 34,35,38,39,40				
N22	A5-6 OUT DISABLE		24	4,7,8,11,13,14,15,16,19,20, 21,22,23,25,26,27,28,29,31, 34,35,38,39,40				
N23	B5-6 OUT DISABLE		22	4,7,8,11,13,14,15,16,19,20, 21,23,24,25,26,27,28,29,31, 34,35,38,39,40				
N24	B5-6 OUT DISABLE		21	4,7,8,11,13,14,15,16,19,20, 22,23,24,25,26,27,28,29,31, 34,35,38,39,40				
N25	MC		20	4,7,8,11,13,14,15,16,19,21, 22,23,24,25,26,27,28,29,31, 34,35,38,39,40		120		

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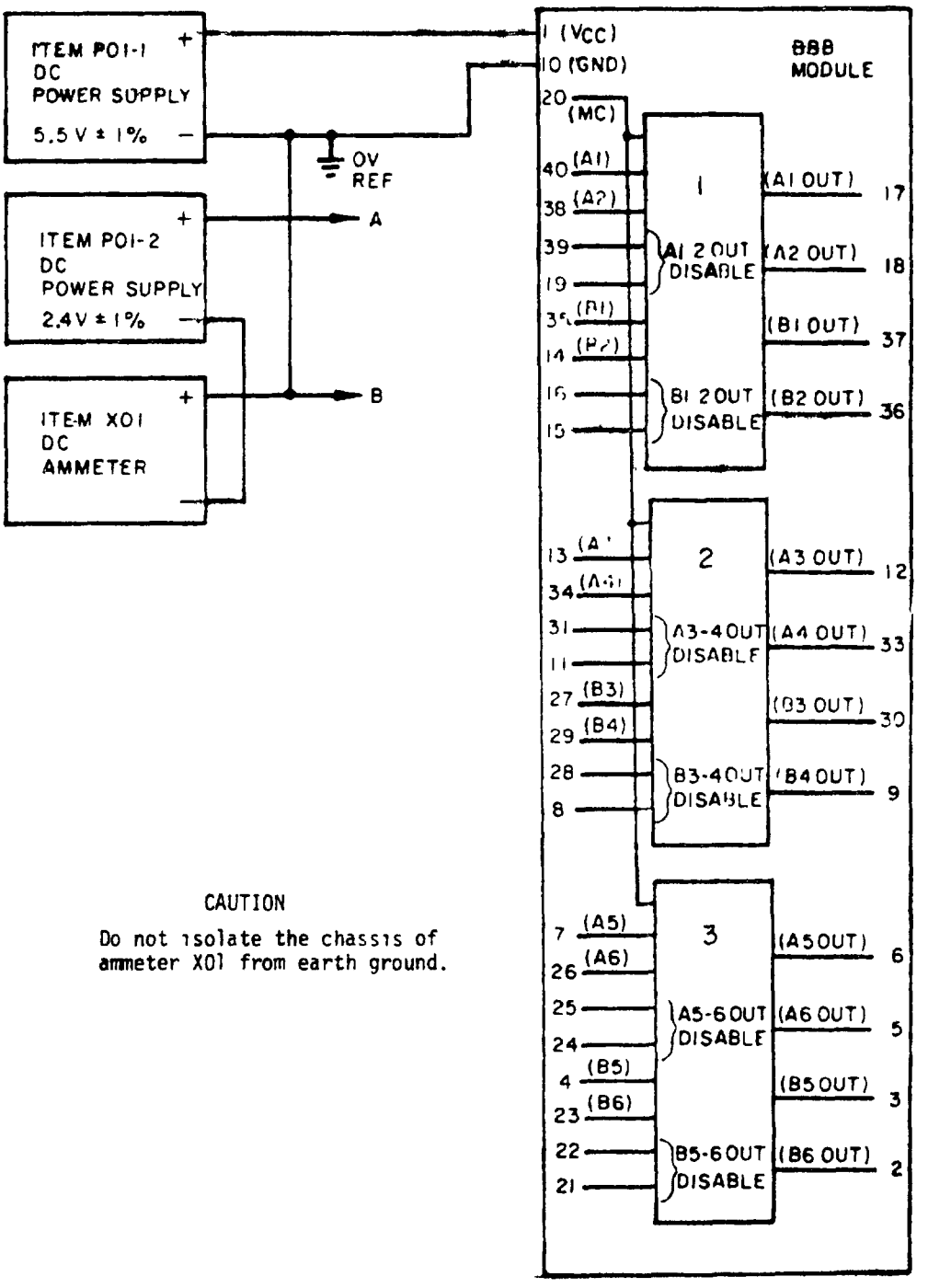


FIGURE 15 Input current (high-level) test circuit for table XII.

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TABLE XIII. Input current (low-level) test.

Test condition	Parameter		Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	Min	Max		
P1	A1	Input current, low level	40	-0.1	-1.73	mA	16
P2	A2		38				
P3	B1		35				
P4	B2		14				
P5	A1-2 OUT DISABLE		39				
P6	A1-2 OUT DISABLE		19				
P7	B1-2 OUT DISABLE		16				
P8	B1-2 OUT DISABLE		15				
P9	A3		13				
P10	A4		34				
P11	B3		27				
P12	B4		29				
P13	A3-4 OUT DISABLE		31				
P14	A3-4 OUT DISABLE		11				
P15	B3-4 OUT DISABLE		28				

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TABLE XIII Input current (low-level) test - Continued.

Test condition	Parameter		Connect A to pin	Initial limits 25° ±5°C Ambient Min	Initial limits 25° ±5°C Ambient Max	Unit	Test figure
P16	B3-4 OUT DISABLE	Input current, low level	8	-0.1	-1.73	mA	16
P17	A5		7				
P18	A6		26				
P19	B5		4				
P20	B6		23				
P21	A5-6 OUT DISABLE		25				
P22	A5-6 OUT DISABLE		24				
P23	B5-6 OUT DISABLE		22				
P24	B5-6 OUT DISABLE		21				
P25	MC		20	-0.3	-5.19		

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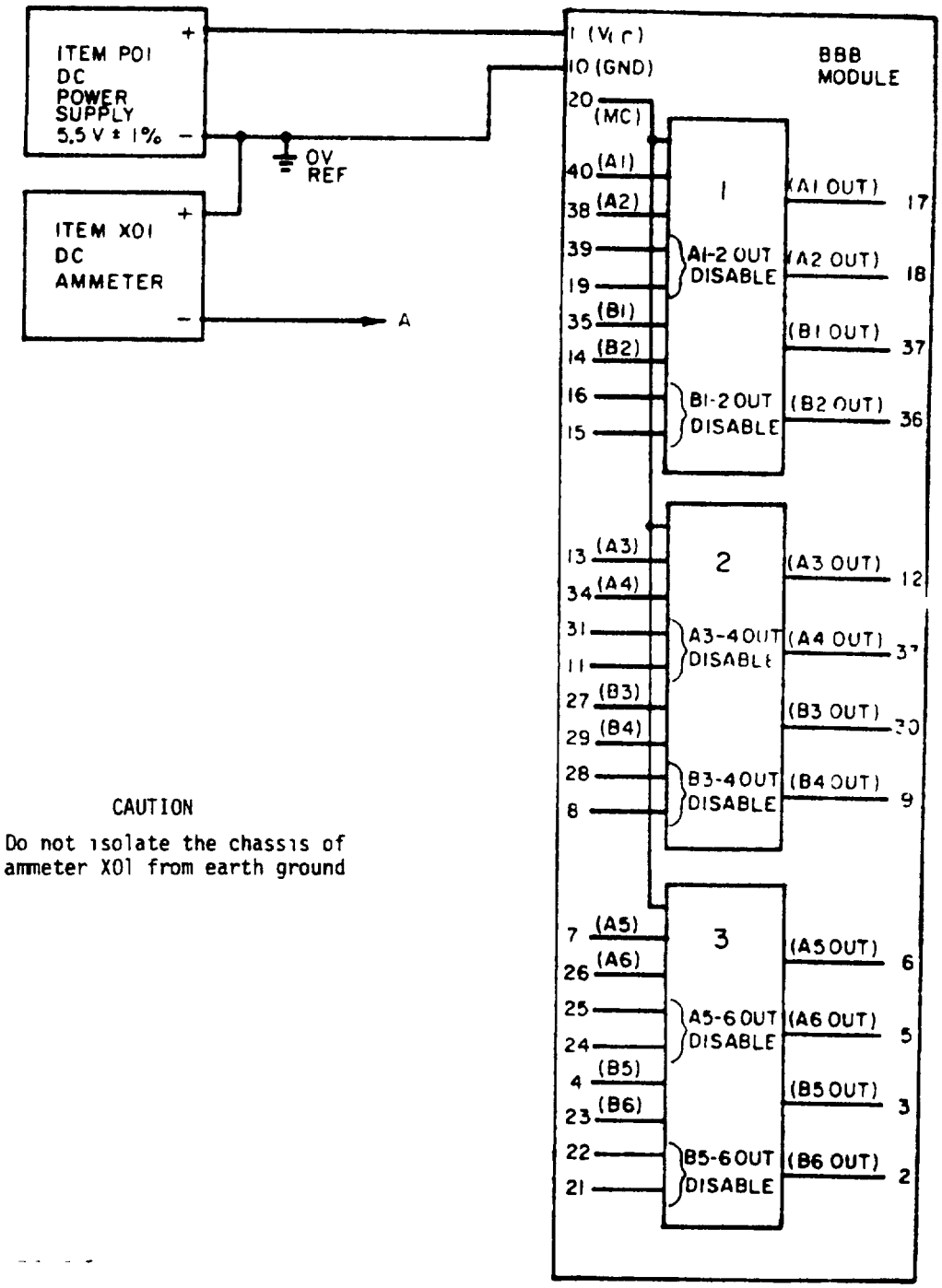


FIGURE 16. Input current (low-level) test circuit for table XIII

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TABLE XIV. Power supply current test

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
Q	Power supply current	50	270	mA	17

TABLE XV Output transition time (t_{TLH} and t_{THL}) test.

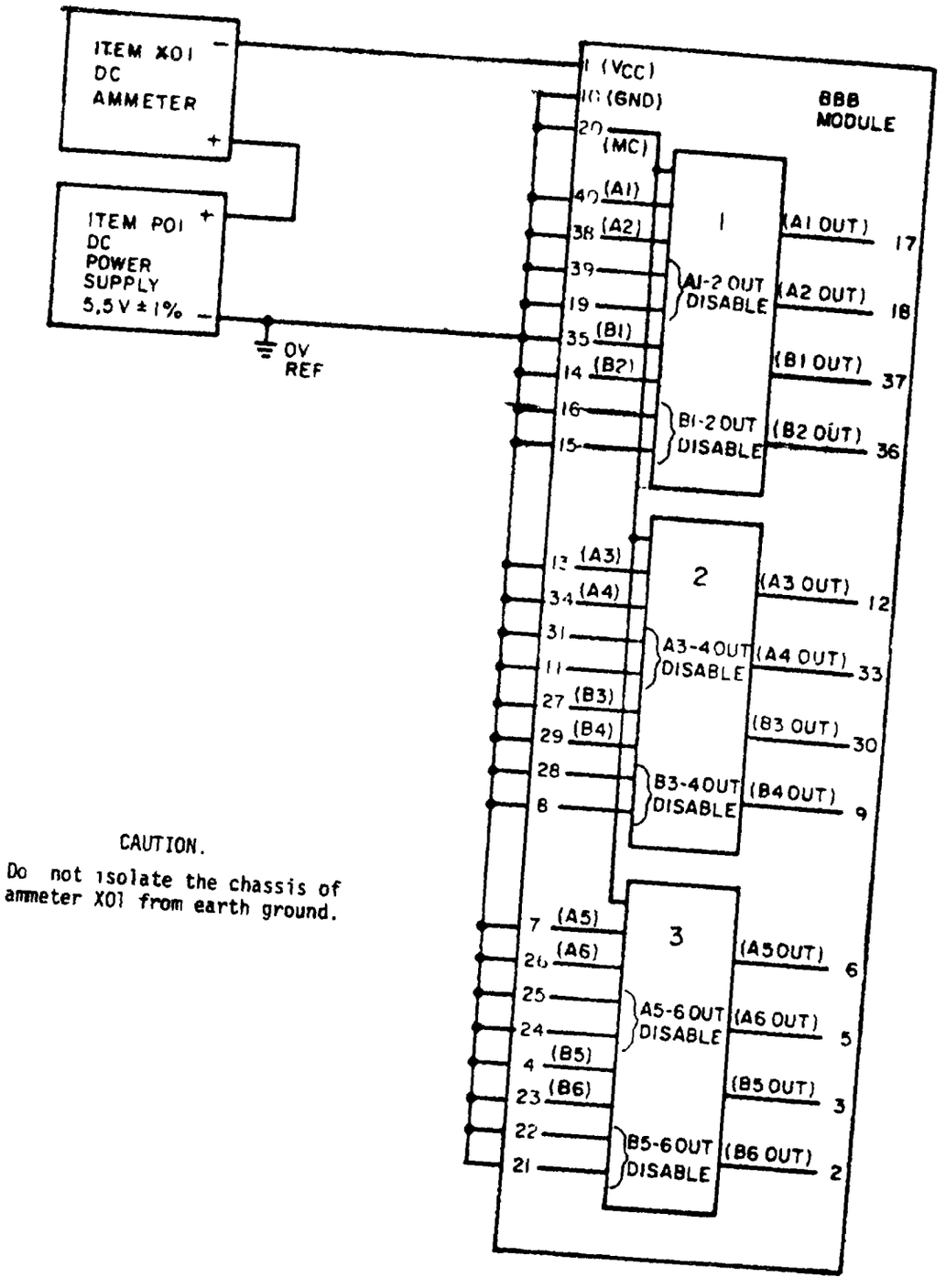
Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
R1	A1 OUT Transition time (t_{TLH}), C_L = 100 pF	40	19,39	17	4	25	ns	18, 19
R2	A2 OUT	38	19,39	18				
R3	B1 OUT	35	15,16	37				
R4	B2 OUT	14	15,16	36				
R5	A3 OUT	13	11,31	12				
R6	A4 OUT	34	11,31	33				
R7	B3 OUT	27	8,28	30				
R8	B4 OUT	29	8,28	9				
R9	A5 OUT	7	24,25	6				
R10	A6 OUT	26	24,25	5				
R11	B5 OUT	4	21,22	3				
R12	B6 OUT	23	21,22	2				

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TABLE XV Output transition time (t_{TLH} and t_{THL}) test - Continued

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
S1	A1 OUT Transition time (t_{THL}), C_L = 100 pF	40	19,39	17	3	20	ns	18, 19
S2	A2 OUT	38	19,39	18				
S3	B1 OUT	35	15,16	37				
S4	B2 OUT	14	15,16	36				
S5	A3 OUT	13	11,31	12				
S6	A4 OUT	34	11,31	33				
S7	B3 OUT	27	8,28	30				
S8	B4 OUT	29	8,28	9				
S9	A5 OUT	7	24,25	6				
S10	A6 OUT	26	24,25	5				
S11	B5 OUT	4	21,22	3				
S12	B6 OUT	23	21,22	2				

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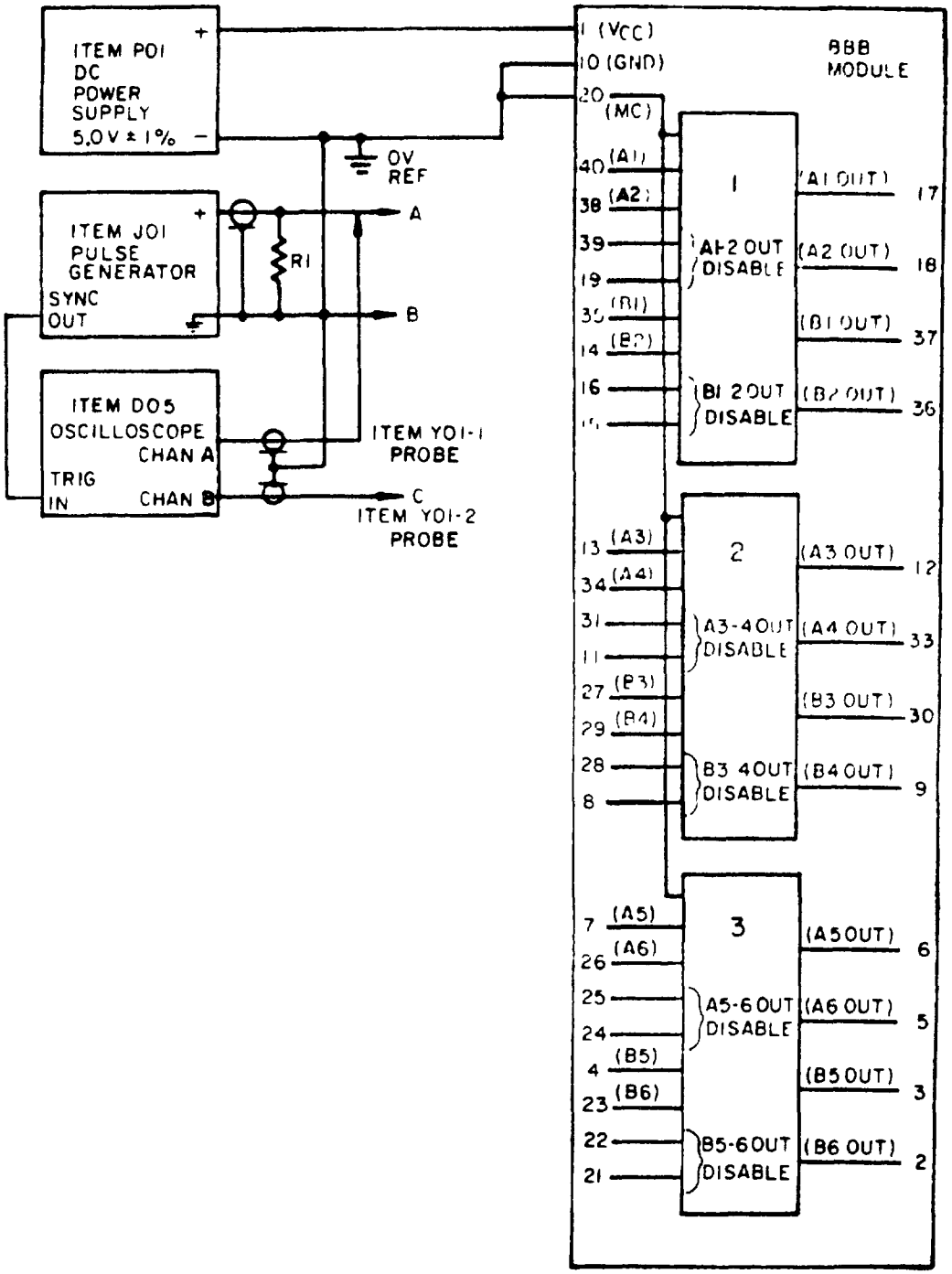


CAUTION.

Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 17. Power supply current test circuit for table XIV.

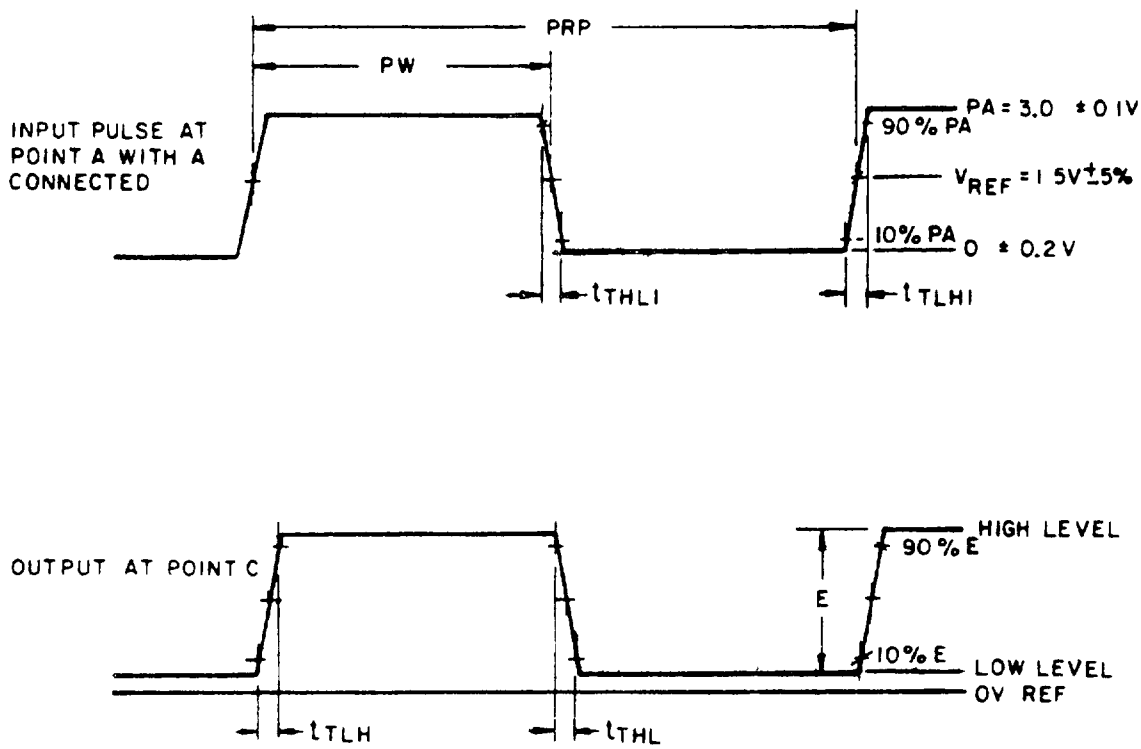
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NOTES
1. R1 = 51.1Ω ±1%, 1/4 W.
2. Load output under test with load as specified on figure 14.

FIGURE 18. Output transition time (t_{TLH} and t_{THL}) test circuit for table XV

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NOTES
1. Input pulse setup:
   PW = 500 ±50 ns.
   PRP = 1,000 ±100 ns
   $t_{TLH1} = t_{THL1}$ = 10 ±2 ns
2. Before making measurements, verify that the output high level is ≥ 2.4 V and the low level is ≤ 0.4 V
3. All voltages levels are referenced to ground.

FIGURE 19. Output transition time (t_{TLH} and t_{THL}) test circuit waveforms for table XV.

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TABLE XVI. Propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
T1	A1 to A1 OUT Propagation delay time (t_{PLH}), C_L = 100 pF	40	20	17	5	25	ns	20, 21
T2	A2 to A2 OUT	38		18				
T3	B1 to B1 OUT	35		37				
T4	B2 to B2 OUT	14		36				
T5	A3 to A3 OUT	13		12				
T6	A4 to A4 OUT	34		33				
T7	B3 to B3 OUT	27		30				
T8	B4 to B4 OUT	29		9				
T9	A5 to A5 OUT	7		6				
T10	A6 to A6 OUT	26		5				
T11	B5 to B5 OUT	4		3				
T12	B6 to B6 OUT	23		2				

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TABLE XVI Propagation delay time (t_{PLH} and t_{PHL}) test - Continued

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
T13	MC to A1 OUT Propagation delay time (t_{PLH}), C_L = 100 pF	20	40	17	5	25	ns	20, 21
T14	MC to B1 OUT		35	37				
T15	MC to A3 OUT	20	13	12				
T16	MC to B3 OUT	20	27	30				
T17	MC to A5 OUT	20	7	6				
T18	MC to B5 OUT	20	4	3				
T19	A1 to A1 OUT Propagation delay time (t_{PLH}), C_L = 1,000 pF	40	20	17	10	50		
T20	A2 to A2 OUT	38		18				
T21	B1 to B1 OUT	35		37				
T22	B2 to B2 OUT	14		36				
T23	A3 to A3 OUT	13		12				
T24	A4 to A4 OUT	34		33				

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TABLE XVI Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
T25	B3 to B3 OUT Propagation delay time (t_{PLH}), C_L = 1,000 pF	27	20	30	10	50	ns	20, 21
T26	B4 to B4 OUT	29		9				
T27	A5 to A5 OUT	7		6				
T28	A5 to A6 OUT	26		5				
T29	B5 to B5 OUT	4		3				
T30	B6 to B6 OUT	23		2				
T31	MC to A1 OUT	20	40	17				
T32	MC to B1 OUT		35	37				
T33	MC to A3 OUT		13	12				
T34	MC to B3 OUT		27	30				
T35	MC to A5 OUT		7	6				
T36	MC to B5 OUT		4	3				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:42Z
Check the source to verify that this is the current version before use.

TABLE XVI Propagation delay time (t_{PLH} and t_{PHL}) test - Continued

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
U1	A1 to A1 OUT Propagation delay time (t_{PHL}), C_L = 100 pF	40	20	17	5	25	ns	20, 21
U2	A2 to A2 OUT	38		18				
U3	B1 to B1 OUT	35		37				
U4	B2 to B2 OUT	14		36				
U5	A3 to A3 OUT	13		12				
U6	A4 to A4 OUT	34		33				
[illegible]	B3 [illegible] OUT	27		30				
U8	B4 to B4 OUT	29		9				
U9	A5 to A5 OUT	7		6				
U10	A6 to A6 OUT	26		5				
U11	B5 to B5 OUT	4		3				
U12	B6 to B6 OUT	23		2				

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TABLE XVI Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

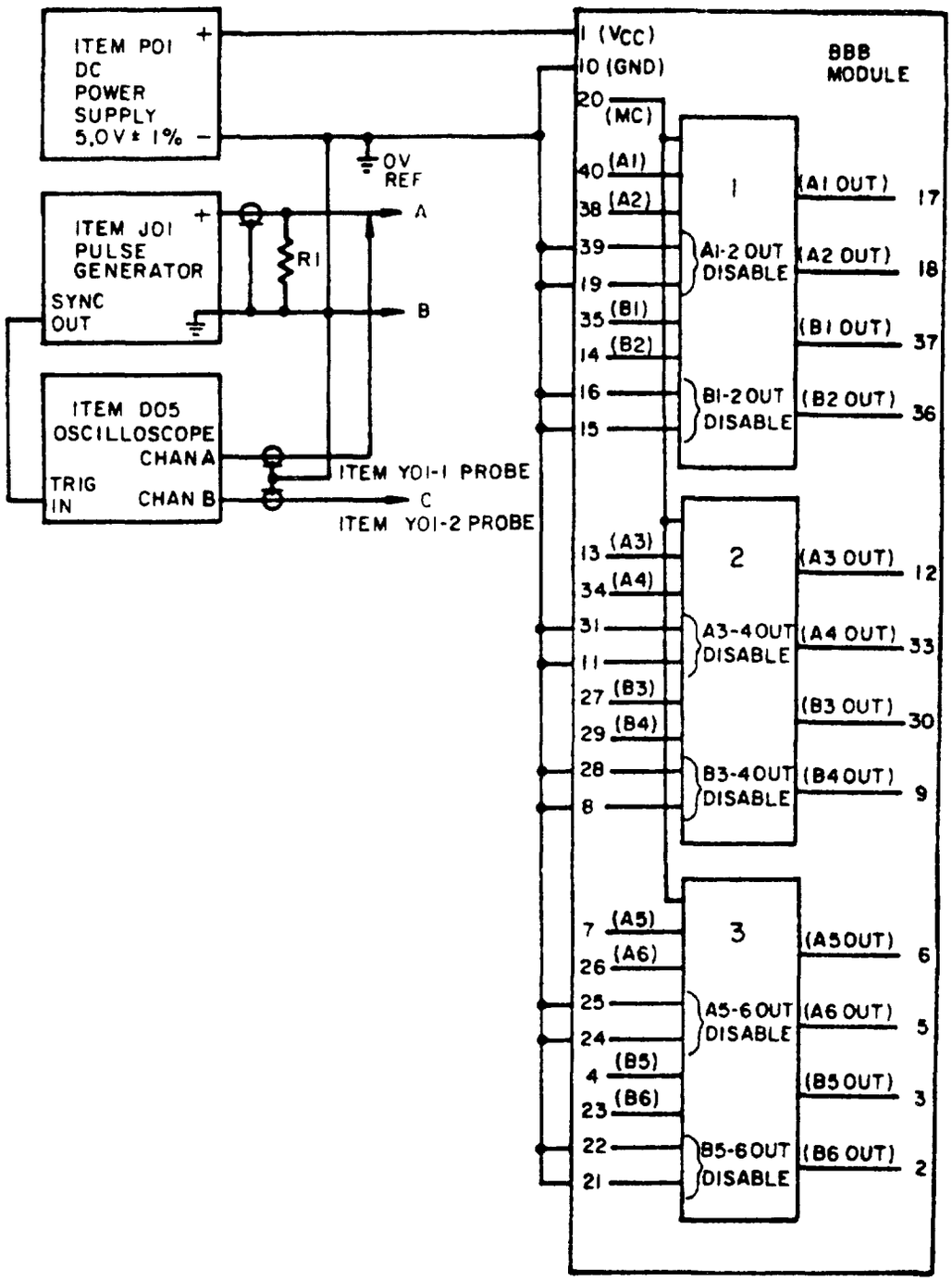
Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
U13	MC to A1 OUT Propagation delay time (t_{PHL}), C_L = 100 pF	20	40	17	5	25	ns	20, 21
U14	MC to B1 OUT		35	37				
U15	MC to A3 OUT		13	12				
U16	MC to B3 OUT		27	30				
U1	MC to A5 OUT		7	6				
U18	MC to B5 OUT		4	3				
U19	A1 to A1 OUT Propagation delay time (t_{PHL}), C_L = 1,000 pF	40	20	17	10	50		
U20	A2 to A2 OUT	38		18				
U21	B1 to B1 OUT	35		37				
U22	B2 to B2 OUT	14		36				
U23	A3 to A3 OUT	13		12				
U24	A4 to A4 OUT	34		33				

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TABLE XVI Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
U25	B3 to B3 OUT Propagation delay time (t_{PHL}), C_L = 1,000 pF	27	20	30	10	50	ns	20, 21
U26	B4 to B4 OUT	29		9				
U27	A5 to A5 OUT	7		6				
U28	A6 to A6 OUT	26		5				
U29	B5 to B5 OUT	4		3				
U30	B6 to B6 OUT	23		2				
U31	MC to A1 OUT	20	40	17				
U32	MC to B1 OUT		35	37				
U33	MC to A3 OUT		13	12				
U34	MC to B3 OUT		27	30				
U35	MC to A5 OUT		7	6				
U36	MC to B5 OUT		4	3				

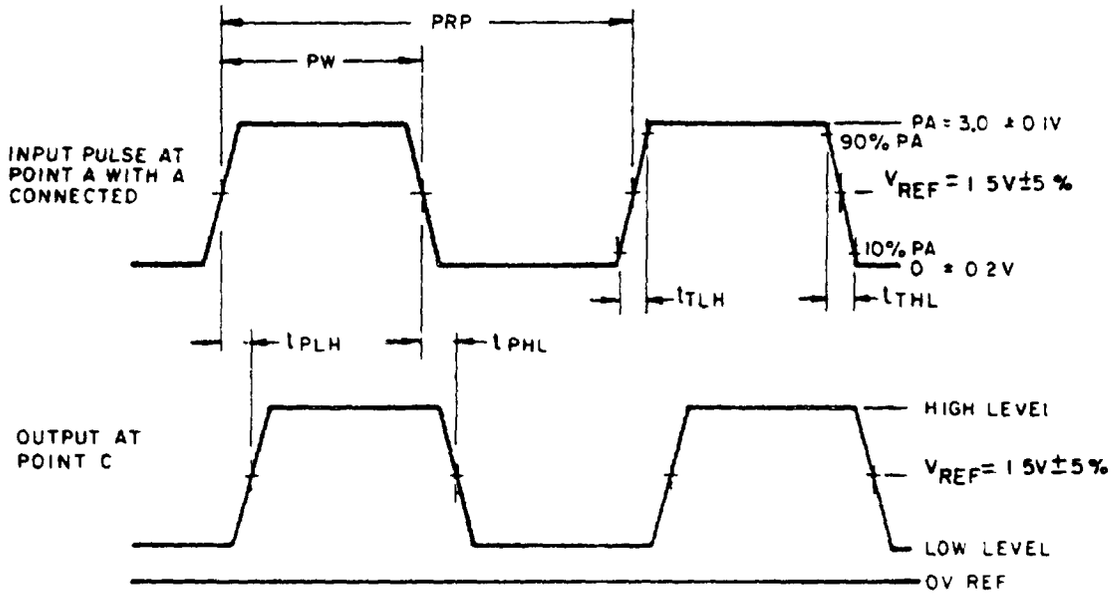
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NOTES:
1. R1 = 51.1Ω ±1%, 1/4 W.
2. Load output under test with load as specified on figure 14.

FIGURE 20. Propagation delay time (t_{PLH} and t_{PHL}) test circuit for table XVI.

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NOTES

1. Input pulse setup:
   PW = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
2. Before making measurements, verify that the output high level is $\geq$ 2.4 V and the low level is $\leq$ 0.4 V.
3. All voltages are referenced to ground.

FIGURE 21. Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table XVI.

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TABLE XVII Disable delay time (T_{oh}, T_{ho}, T_{lh}, and T_{hl}) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
V1	A1 OUT Disable delay time (T_{oh}), C_L = 50 pF	Open	19,40	39	17	5	35	ns	22, 23
V2	A1 OUT		39,40	19	17				
V3	A2 OUT		19,38	39	18				
V4	A2 OUT		38,39	19	18				
V5	B1 OUT		15,35	16	37				
V6	B1 OUT		16,35	15	37				
V7	B2 OUT		14,15	16	36				
V8	B2 OUT		14,16	15	36				
V9	A3 OUT		11,13	31	12				
V10	A3 OUT		13,31	11	12				
V11	A4 OUT		11,34	31	33				
V12	A4 OUT		31,34	11	33				
V13	B3 OUT		8,27	28	30				
V14	B3 OUT		27,28	8	30				
V15	B4 OUT		8,29	28	9				
V16	B4 OUT		28,29	8	9				
V17	A5 OUT		7,24	25	6				
V18	A5 OUT		7,25	24	6				

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TABLE XVII. Disable delay time (T_{oh}, T_{ho}, T_{lh}, and T_{hl}) test - Continued

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
V19	A6 OUT Disable delay time (T_{oh}), C_L = 50 pF	Open	24,26	25	5	5	35	ns	22, 23
V20	A6 OUT		25,26	24	5				
V21	B5 OUT		4,21	22	3				
V22	B5 OUT		4,22	21	3				
V23	B6 OUT		21,23	22	2				
V24	B6 OUT		22,23	21	2				
W1	A1 OUT Disable delay time (T_{ho}), C_L = 50 pF		19,40	39	17				
W2	A1 OUT		39,40	19	17				
W3	A2 OUT		19,38	39	18				
W4	A2 OUT		38,39	19	18				
W5	B1 OUT		15,35	16	37				
W6	B1 OUT		16,35	15	37				
W7	B2 OUT		14,15	16	36				
W8	B2 OUT		14,16	15	36				
W9	A3 OUT		11,13	31	12				
W10	A3 OUT		13,31	11	12				
W11	A4 OUT		11,34	31	33				

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TABLE XVII. Disable delay time (T_{oh}, T_{ho}, T_{lh}, and T_{hl}) test - Continued

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
W12	A4 OUT Disable delay time (T_{ho}), C_L = 50 pF	Open	31,34	11	33	5	35	ns	22, 23
W13	B3 OUT		8,27	28	30				
W14	B3 OUT		27,28	8	30				
W15	B4 OUT		8,29	28	9				
W16	B4 OUT		28,29	8	9				
W17	A5 OUT		7,24	25	6				
W18	A5 OUT		7,25	24	6				
W19	A6 OUT		24,26	25	5				
W20	A6 OUT		25,26	24	5				
W21	B5 OUT		4,21	22	3				
W22	B5 OUT		4,22	21	3				
W23	B6 OUT		21,23	22	2				
W24	B6 OUT		22,23	21	2				
X1	A1 OUT Disable delay time (T_{lh}), C_L = 50 pF	40	19	39	17				
X2	A1 OUT Disable delay time (T_{lh}), C_L = 50 pF	40	39	19	17				
X3	A2 OUT Disable delay time (T_{lh}), C_L = 50 pF	38	19	39	18				

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TABLE XVII Disable delay time (T_{oh}, T_{ho}, T_{lh}, and T_{hl}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
X4	A2 OUT Disable delay time (T_{lh}), C_L = 50 pF	38	39	19	18	5	35	ns	22, 23
X5	B1 OUT	35	15	16	37				
X6	B1 OUT	35	16	15	37				
X7	B2 OUT	14	15	16	36				
X8	B2 OUT	14	16	15	36				
X9	A3 OUT	13	11	31	12				
X10	A3 OUT	13	31	11	12				
X11	A4 OUT	34	11	31	33				
X12	A4 OUT	34	31	11	33				
X13	B3 OUT	27	8	28	30				
X14	B3 OUT	27	28	8	30				
X15	B4 OUT	29	8	28	9				
X16	B4 OUT	29	28	8	9				
X17	A5 OUT	7	24	25	6				
X18	A5 OUT	7	25	24	6				
X19	A6 OUT	26	24	25	5				
X20	A6 OUT	26	25	24	5				

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TABLE XVII Disable delay time (T_{oh}, T_{ho}, T_{lh}, and T_{hl}) test. - Continued

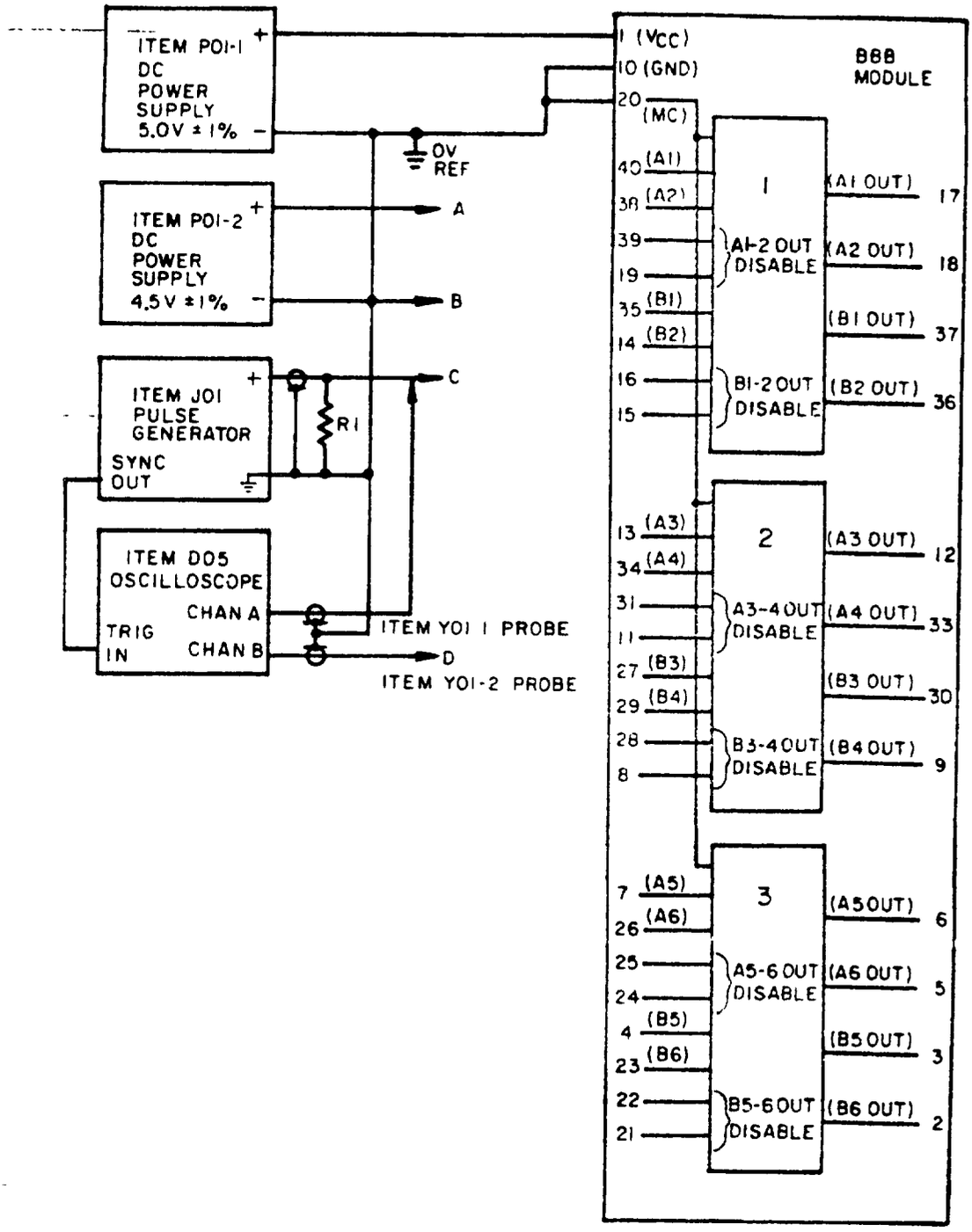
Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
X21	B5 OUT Disable delay time (T_{lh}), C_L = 50 pF	4	21	22	3	5	35	ns	22, 23
X22	B5 OUT	4	22	21	3				
X23	B6 OUT	23	21	22	2				
X24	B6 OUT	23	22	21	2				
Y1	A1 OUT Disable delay time (T_{hl}), C_L = 50 pF	40	19	39	17				
Y2	A1 OUT	40	20	19	17				
Y3	A2 OUT	38	19	39	18				
Y4	A2 OUT	38	39	19	18				
Y5	B1 OUT	35	15	16	37				
Y6	B1 OUT	35	16	15	37				
Y7	B2 OUT	14	15	16	36				
Y8	B2 OUT	14	16	15	36				
Y9	A3 OUT	13	11	31	12				
Y10	A3 OUT	13	31	11	12				
Y11	A4 OUT	34	11	31	33				
Y12	A4 OUT	34	31	11	33				
Y13	B3 OUT	27	8	28	30				

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TABLE XVII. Disable delay time (T_{oh}, T_{ho}, T_{lh}, and T_{hl}) test - Continued

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Y14	B3 OUT Disable delay time (T_{hl}), C_L = 50 pF	27	28	8	30	5	35	ns	22, 23
Y15	B4 OUT	29	8	28	9				
Y16	B4 OUT	29	28	8	9				
Y17	A5 OUT	7	24	25	6				
Y18	A5 OUT	7	25	24	6				
	OUT	26	24	25	5				
Y20	A6 OUT	26	25	24	5				
Y21	B5 OUT	4	21	22	3				
Y22	B5 OUT	4	22	21	3				
Y23	B6 OUT	23	21	22	2				
Y24	B6 OUT	23	22	21	2				

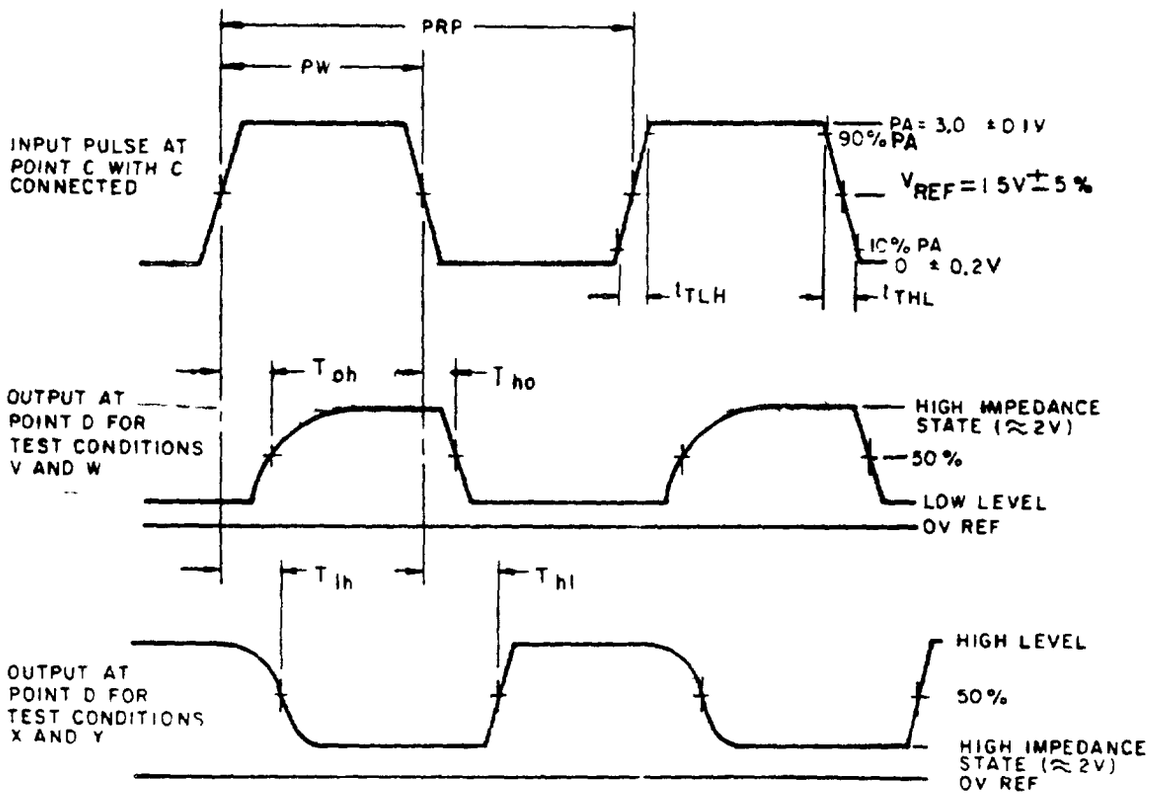
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NOTES:
1. R1 = 51.1Ω ±1%, 1/4 W.
2. Load the output under test with the load as specified on figure 14

FIGURE 22 Disable delay time (T_{oh}, T_{ho}, T_{lh}, and T_{hl}) test circuit for table XVII

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NOTES

1. Input pulse setup.
   PW = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns
2. Before making measurements, verify that the output high level is ≥ 2.4 V and the low level is ≤ 0.4 V.
3. All voltage levels are referenced to ground.

FIGURE 23. Disable delay time (T_{oh}, T_{ho}, T_{lh}, and T_{hl}) test circuit waveforms for table XVII

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TABLE XVIII. Differential delay time (t_{PLH} and t_{PHL}) test.

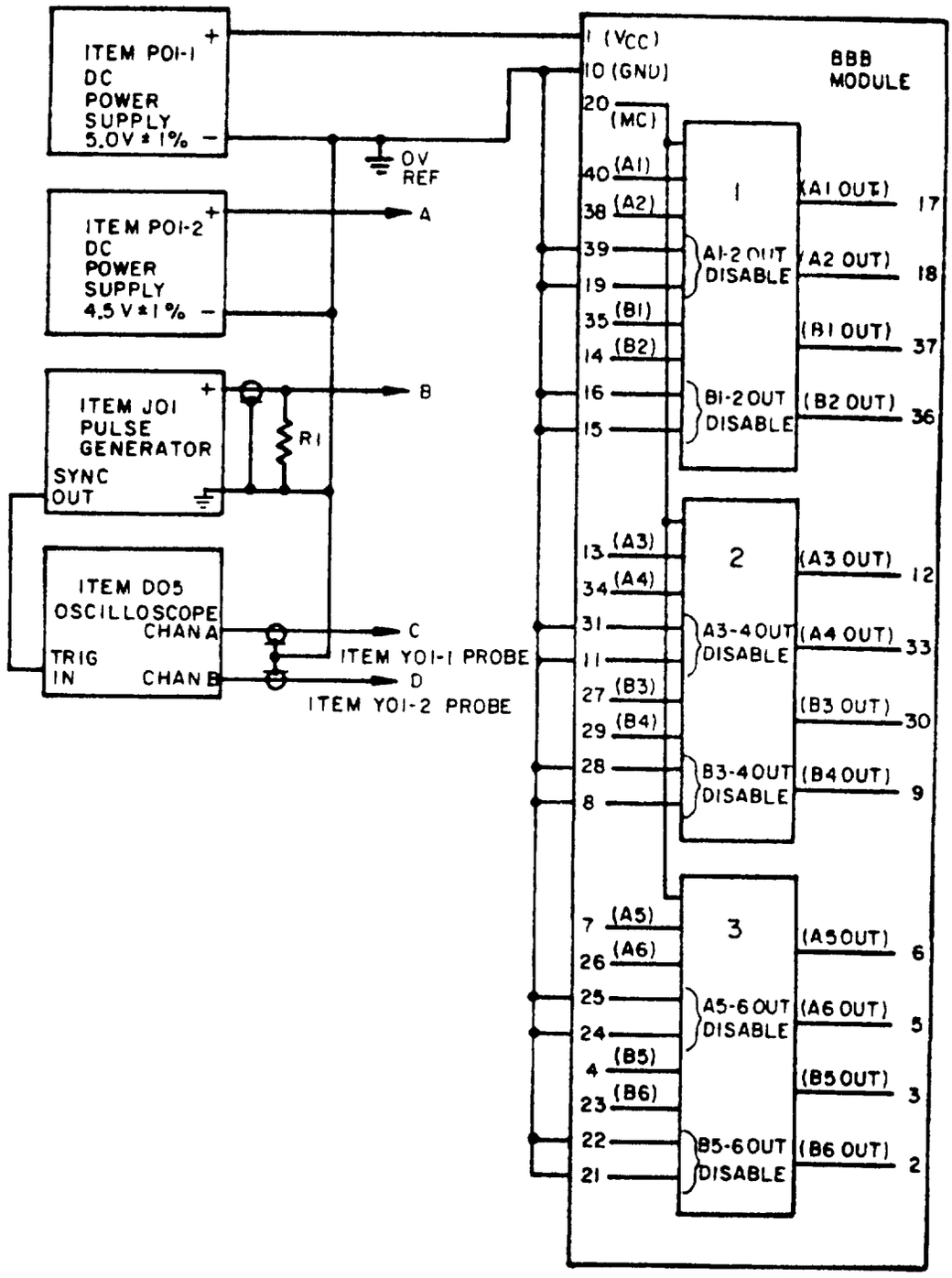
Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Z1	A1 OUT to A2 OUT Differential delay time (Δt_{PLH}), C_L = 50 pF	20	38,40	17	18	-10	10	ns	24,25
Z2	B1 OUT to B2 OUT		14,35	37	36				
Z3	A3 OUT to A4 OUT		13,34	12	33				
Z4	B3 OUT to B4 OUT		27,29	30	9				
Z5	A5 OUT to A6 OUT		7,26	6	5				
Z6	B5 OUT to B6 OUT		4,23	3	2				
Z7	A1 OUT to A2 OUT Differential delay time (Δt_{PHL}), C_L = 50 pF		38,40	17	18				
Z8	B1 OUT to B2 OUT Differential delay time (Δt_{PHL}), C_L = 50 pF		14,35	37	36				
Z9	A3 OUT to A4 OUT Differential delay time (Δt_{PHL}), C_L = 50 pF		13,34	12	33				

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TABLE XVIII. Differential delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Z10	B3 OUT Differential delay time to (Δt_{PHL}), B4 OUT C_L = 50 pF	20	27,29	30	9	-10	10	ns	24,25
Z11	A5 OUT Differential delay time to (Δt_{PHL}), A6 OUT C_L = 50 pF	20	7,26	6	5	-10	10	ns	24,25
Z12	B5 OUT Differential delay time to (Δt_{PHL}), B6 OUT C_L = 50 pF	20	4,23	3	2	-10	10	ns	24,25

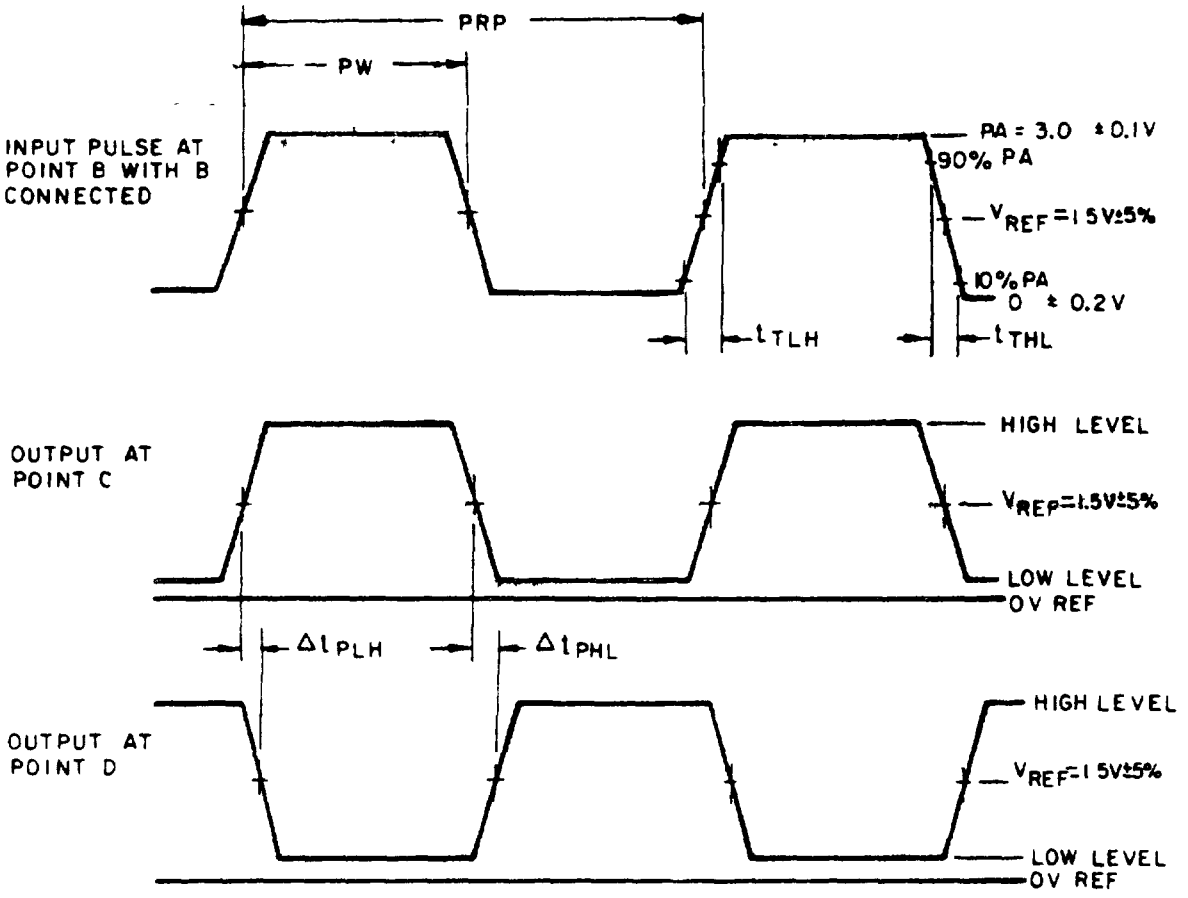
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NOTES:
1. R1 = 51 1Ω ±1%, 1/4 W.
2 Load the two outputs under test with the load as specified on figure 14.

FIGURE 24 Differential delay time ($\Delta tPLH$ and $\Delta tPHL$) test circuit for table XVIII

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NOTES.
1. Input pulse setup:
   PW = 500 ±50 ns
   PRP = 1,000 ±100 ns
   $t_{TLH} = t_{THL}$ = 10 ±2 ns
2. Before making measurements, verify that the output high level is $\geq$ 2.4 V and the low level is $\leq$ 0.4 V.
3. All voltage levels are referenced to ground.

FIGURE 25. Differential delay time (Δt_{PLH} and Δt_{PHL}) test circuit waveforms for table XVIII.

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TABLE XIX. Output crosstalk (single-ended mode) (high-level and low-level) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
AA1	A1 OUT Output crosstalk, high level (single-ended mode)	40	Open	4,7,13,14,23,26, 27,29,34,35,38	17	0	700	mV	26,27
AA2	A2 OUT	38		4,7,13,14,23,26, 27,29,34,35,40	18				
AA3	B1 OUT	35		4,7,13,14,23,26, 27,29,34,38,40	37				
AA4	B2 OUT	14		4,7,13,23,26,27, 29,34,35,38,40	36				
AA5	A3 OUT	13		4,7,14,23,26,27, 29,34,38,40	12				
AA6	A4 OUT	34		4,7,13,14,23,26, 27,29,35,38,40	33				
AA7	B3 OUT	27		4,7,13,14,23,26, 29,34,35,38,40	30				
AA8	B4 OUT	29		4,7,13,14,23,26, 27,34,35,38,40	9				
AA9	A5 OUT	7		4,13,14,23,26,27, 29,34,35,38,40	6				
AA10	A6 OUT	26		4,7,13,14,23,27, 29,34,35,38,40	5				
AA11	B5 OUT	4		7,13,14,23,26,27, 29,34,35,38,40	3				
AA12	B6 OUT	23		4,7,13,14,26,27, 29,34,35,38,40	2				

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TABLE XIX Output crosstalk (single-ended mode) (high-level and low-level) test - Continued

Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
AA13	A1 OUT	Output crosstalk, high level (single-ended) mode	14,35, 38,40	Open	4,7,13,23,26,27, 29,34	17	0	550	mV	26,27
AA14	A2 OUT					18				
AA15	B1 OUT					37				
AA16	B2 OUT					36				
AA17	A3 OUT		13,27, 29,34		4,7,14,23,26,35, 38,40	12				
AA18	A4 OUT					33				
AA19	B3 OUT					30				
AA20	B4 OUT					9				
AA21	A5 OUT		4,7, 23,26		13,14,27,29,34, 35,38,40	6				
AA22	A6 OUT					5				
AA23	B5 OUT					3				
AA24	B6 OUT					2				
BB1	A1 OUT	Output crosstalk, low level (single-ended mode	Open	40	4,7,13,14,23,26, 27,29,34,35,38	17	0.0	1.49	V	
BB2	A2 OUT	Output crosstalk, low level (single-ended mode	Open	38	4,7,13,14,23,26, 27,29,34,35,40	18	0.0	1.49	V	

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TABLE XIX. Output crosstalk (single-ended mode) (high-level and low-level) test - Continued.

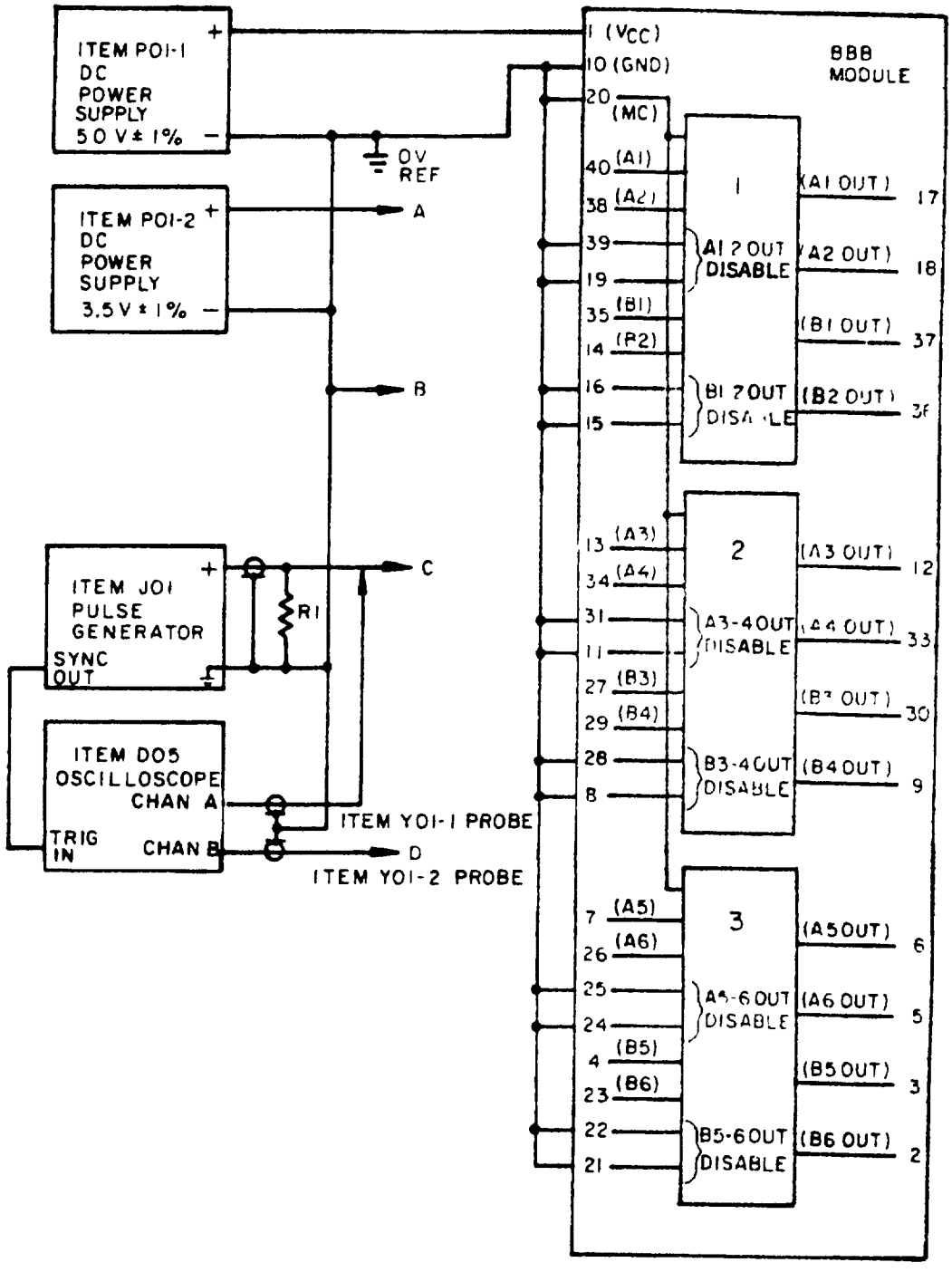
Test condition	Parameter		Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
BB3	B1 OUT	Output crosstalk, low level (single-ended mode)	Open	35	4,7,13,14,23,26, 27,29,34,38,40	37	0.0	1.49	V	26,27
BB4	B2 OUT			14	4,7,13,23,26,27, 29,34,35,38,40	36				
BB5	A3 OUT			13	4,7,14,23,26,27, 29,34,35,38,40	12				
BB6	A4 OUT			34	4,7,13,14,23,26, 27,29,35,38,40	33				
BB7	B3 OUT			27	4,7,13,14,23,26, 29,34,35,38,40	30				
BB8	B4 OUT			29	4,7,13,14,23,26, 27,34,35,38,40	9				
BB9	A5 OUT			7	4,13,14,23,26, 27,29,34	6				
BB10	A6 OUT			26	4,7,13,14,23,27, 29,34,35,38,40	5				
BB11	B5 OUT			4	7,13,14,23,26, 27,29,34,35,38, 40	3				
BB12	B6 OUT			23	4,7,13,14,26,27, 29,34,35,38,40	2				
BB13	A1 OUT			14,35, 38,40	4,7,13,23,26,27, 29,34	17	0	560	mV	

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TABLE XIX. Output crosstalk (single-ended mode) (high-level and low-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
BB14	A2 OUT Output crosstalk, low level (single-ended mode)	Open	14,35, 38,40	4,7,13,23,26,27, 29,34	18	0	560	mV	26,27
BB15	B1 OUT				37				
BB16	B2 OUT				36				
BB17	A3 OUT		13,27, 29,34	4,7,14,23,26,35, 38,40	12				
BB18	A4 OUT				33				
BB19	B3 OUT				30				
BB20	B4 OUT				9				
BB21	A5 OUT		4,7, 23,26	13,14,27,29,34, 35,38,40	6				
BB22	A6 OUT				5				
BB23	B5 OUT				3				
BB24	B6 OUT				2				

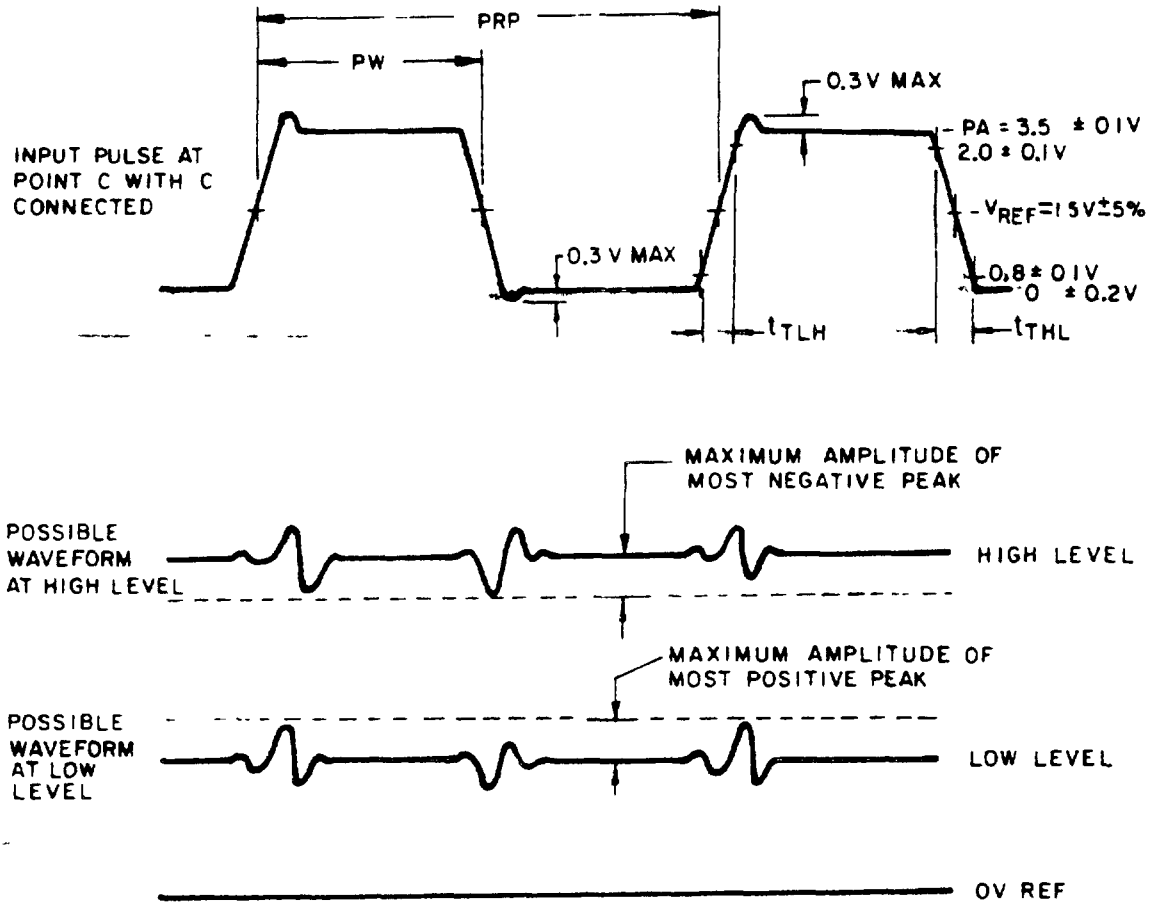
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NOTES:
1 R1 = 51.1Ω ±1%, 1/4 W
2 Load all the outputs with the load as specified on figure 14.

FIGURE 26 Output crosstalk (single-ended mode) (high-level and low-level) test circuit for table XIX

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NOTE. Input pulse setup:
PW = 500 ±50 ns.
PRP = 1,000 ± 100 ns.
$t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at each input pin).

FIGURE 27. Output crosstalk (single-ended mode) (high-level and low-level) test circuit waveforms for table XIX.

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TABLE XX Output crosstalk (differential mode) test.

Test condition	Parameter	Connect					Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
CC1	A1 OUT to A2 OUT — Output crosstalk (differential mode)	20	38,40	4,7,13,14,23,26,27,29,34,35	17	18	0.0	1.0	V	28,29
CC2	B1 OUT to B2 OUT		14,35	4,7,13,23,26,27,29,34,38,40	37	36				
CC3	A3 OUT to A4 OUT		13,34	4,7,14,23,26,27,29,35,38,40	12	33				
CC4	B3 OUT to B4 OUT		27,29	4,7,13,14,23,26,34,35,38,40	30	9				
CC5	A5 OUT to A6 OUT		7,26	4,13,14,23,27,29,34,35,38,40	6	5				
CC6	B5 OUT to B6 OUT		4,23	7,13,14,26,27,29,34,35,38 40	3	2				
CC7	A2 OUT to A1 OUT	20,38,40	Open	4,7,13,14,23,26,27,29,34,35	18	17				
CC8	B2 OUT to B1 OUT	14,20,35		4,7,13,23,26,27,29,34,38,40	36	37				
CC9	A4 OUT to A3 OUT	13,20,34		4,7,14,23,26,27 29,35,38,40	33	12				

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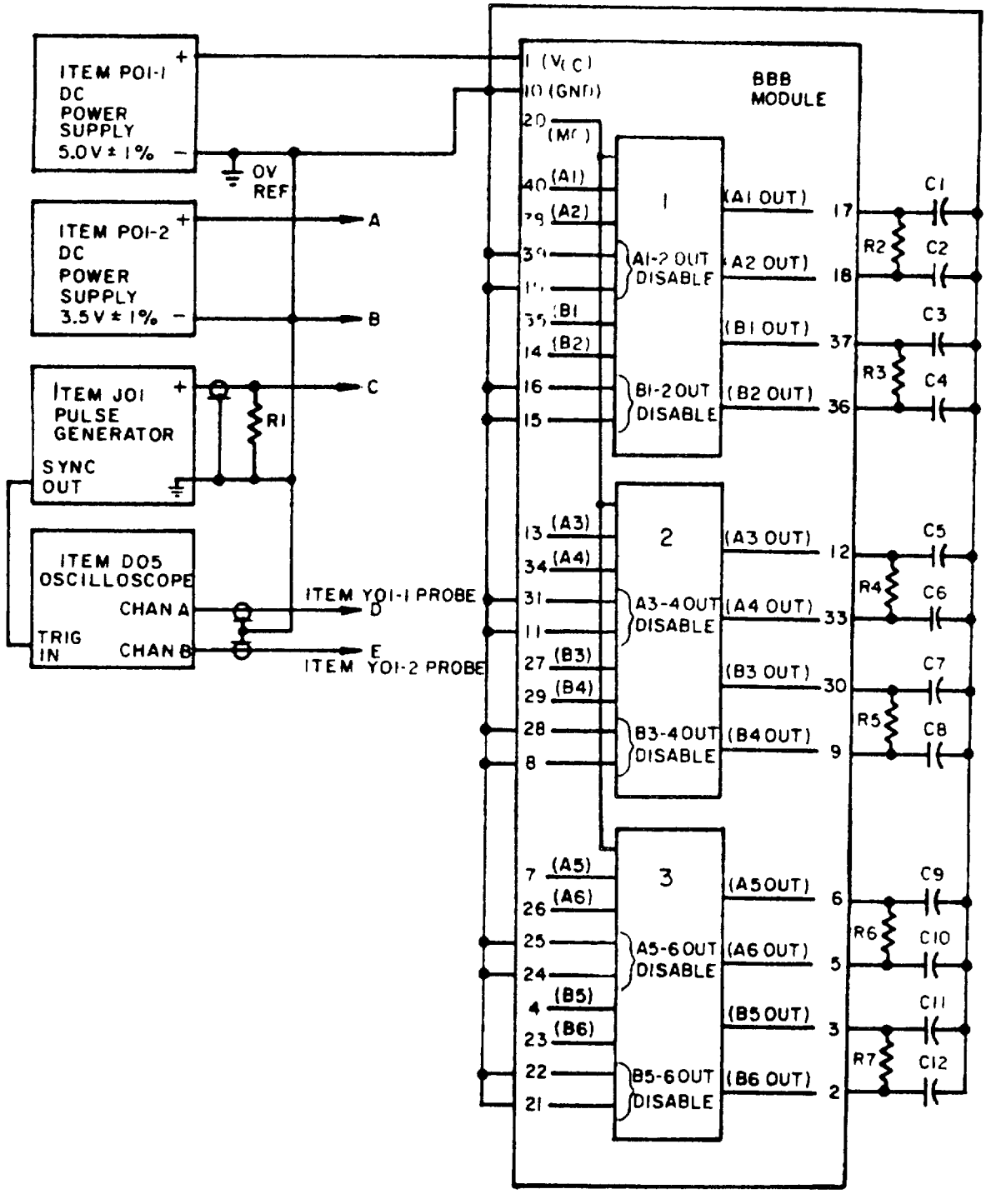
TABLE XX. Output crosstalk (differential mode) test.

Test condition	Parameter	Connect					Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
CC10	B4 OUT to B3 OUT Output crosstalk (differential mode)	20,27, 29	Open	4,7,13,14,23, 26,34,35,38, 40	9	30	0.0	1.0	V	28,29
CC11	A6 OUT to A5 OUT Output crosstalk (differential mode)	7,20, 26	Open	4,13,14,23, 27,29,34,35, 38,40	5	6	0.0	1.0	V	28,29
CC12	B6 OUT to B5 OUT Output crosstalk (differential mode)	4,20, 23	Open	7,13,14,26, 27,29,34,35, 38,40	2	3	0.0	1.0	V	28,29

TABLE XXI. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
DD	Power supply transient amplitude	0	600	mV	30,31

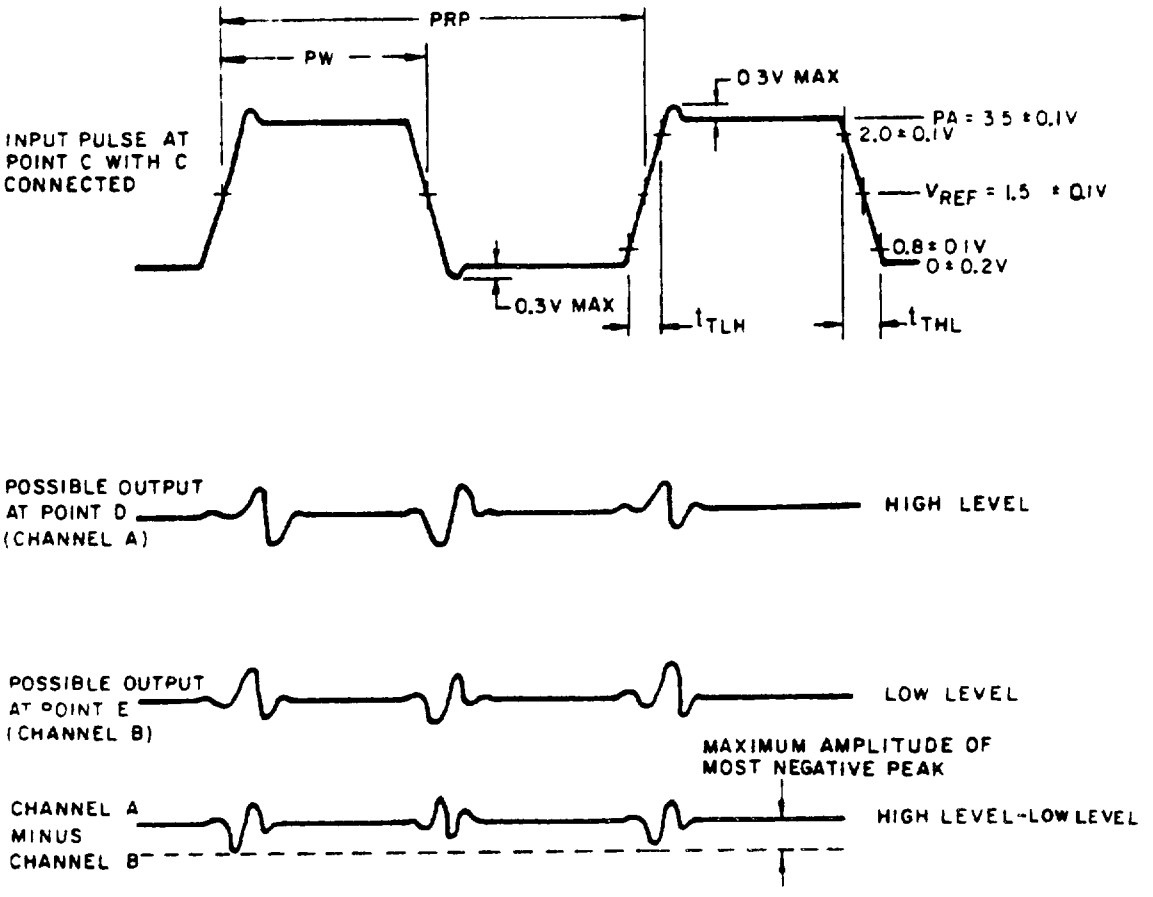
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NOTES:
1. R1 = 51.1Ω ±1%, 1/4 W
2. R2-R7 = 150Ω ±1%, 1/4 W
3 C1-C12 = 50 ±5 pF (including probe and parasitic capacitance)

FIGURE 28 Output crosstalk (differential mode) test circuit for table XX

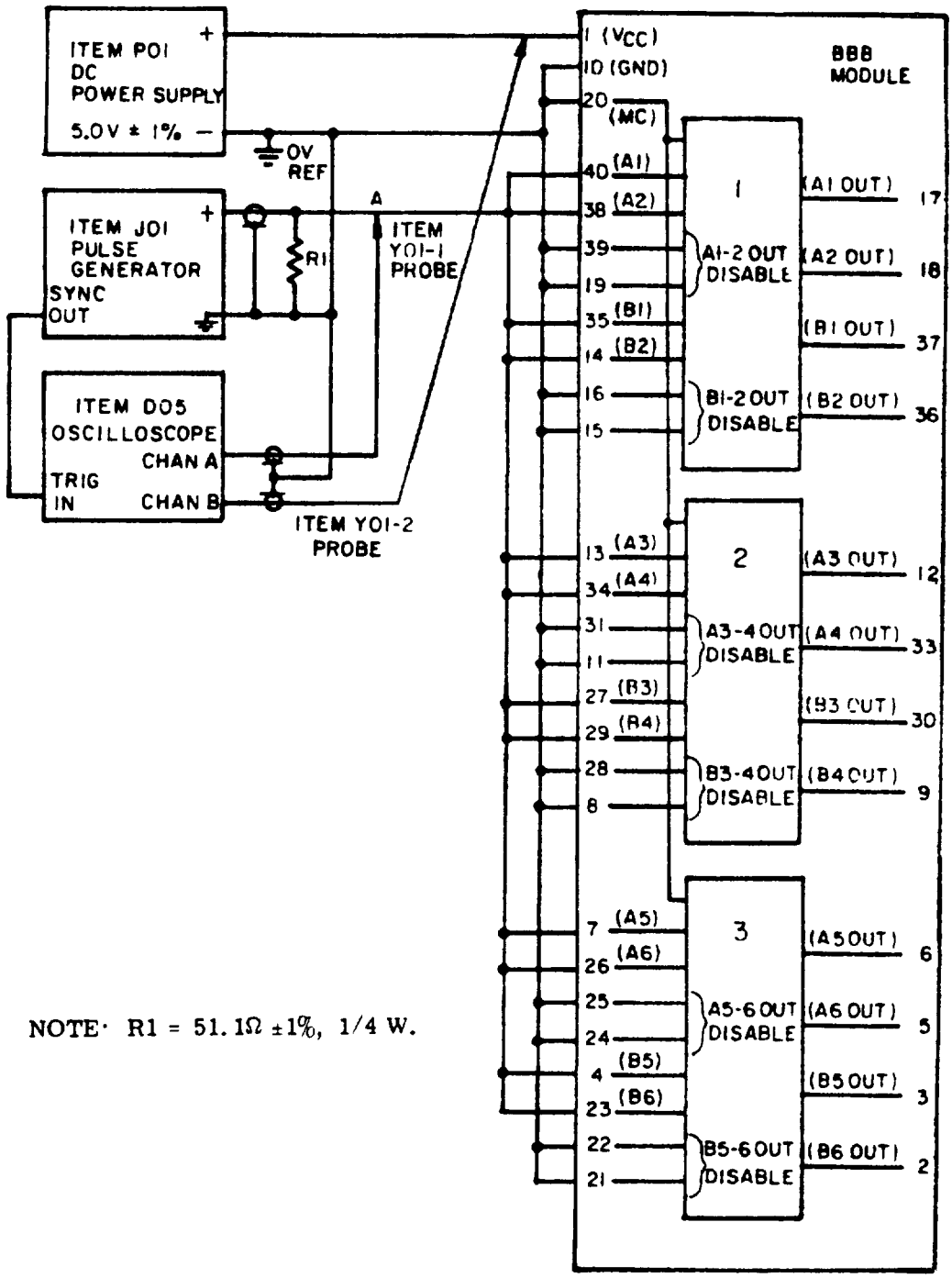
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NOTE Input pulse parameters:
PW = 500 ±50 ns.
PRP = 1,000 ±100 ns.
$t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at each input pin).

FIGURE 29. Output crosstalk (differential mode) test circuit waveforms for table XX.

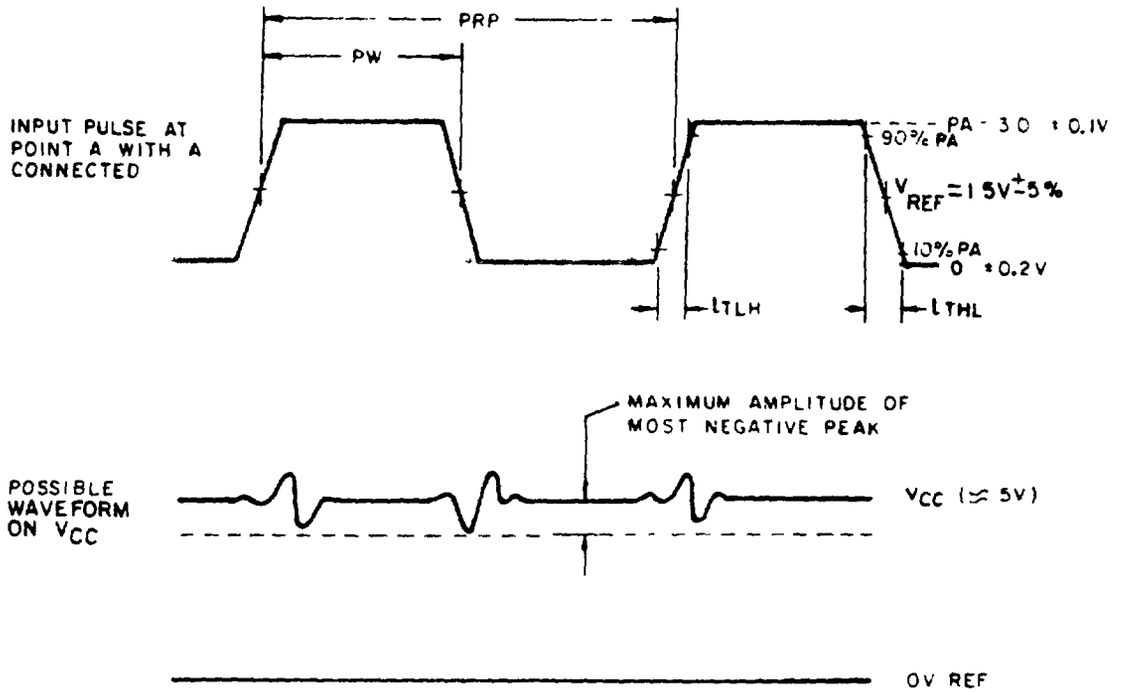
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NOTE: R1 = 51.1Ω ±1%, 1/4 W.

FIGURE 30. Power supply transient amplitude test circuit for table XXI.

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NOTE: Input pulse setup
PW = 500 ±50 ns.
PRP = 1,000 ±100 ns.
$t_{TLH} = t_{THL}$ = 10 ±2 ns.

FIGURE 31 Power supply transient amplitude test circuit waveforms for table XXI.

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TABLE XXII. Power supply filter, input, and output capacitance test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
EE	Power supply filter capacitance	1	Open	1.6	7.2	µF	32
FF1	A1 Input capacitance	Open	40	2	15	pF	
FF2	A2		38				
FF3	B1		35				
FF4	B2		14				
FF5	A1-2 OUT DISABLE		39				
FF6	A1-2 OUT DISABLE		19				
FF7	B1-2 OUT DISABLE		16				
FF8	B1-2 OUT DISABLE		15				
FF9	A3		13				
FF10	A4		34				
FF11	B3		27				
FF12	B4		29				
FF13	A3-4 OUT DISABLE		31				
FF14	A3-4 OUT DISABLE		11				
FF15	B3-4 OUT DISABLE		28				
FF16	B3-4 OUT DISABLE		8				
FF17	A5		7				

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TABLE XXII Power supply filter, input, and output capacitance test - Continued.

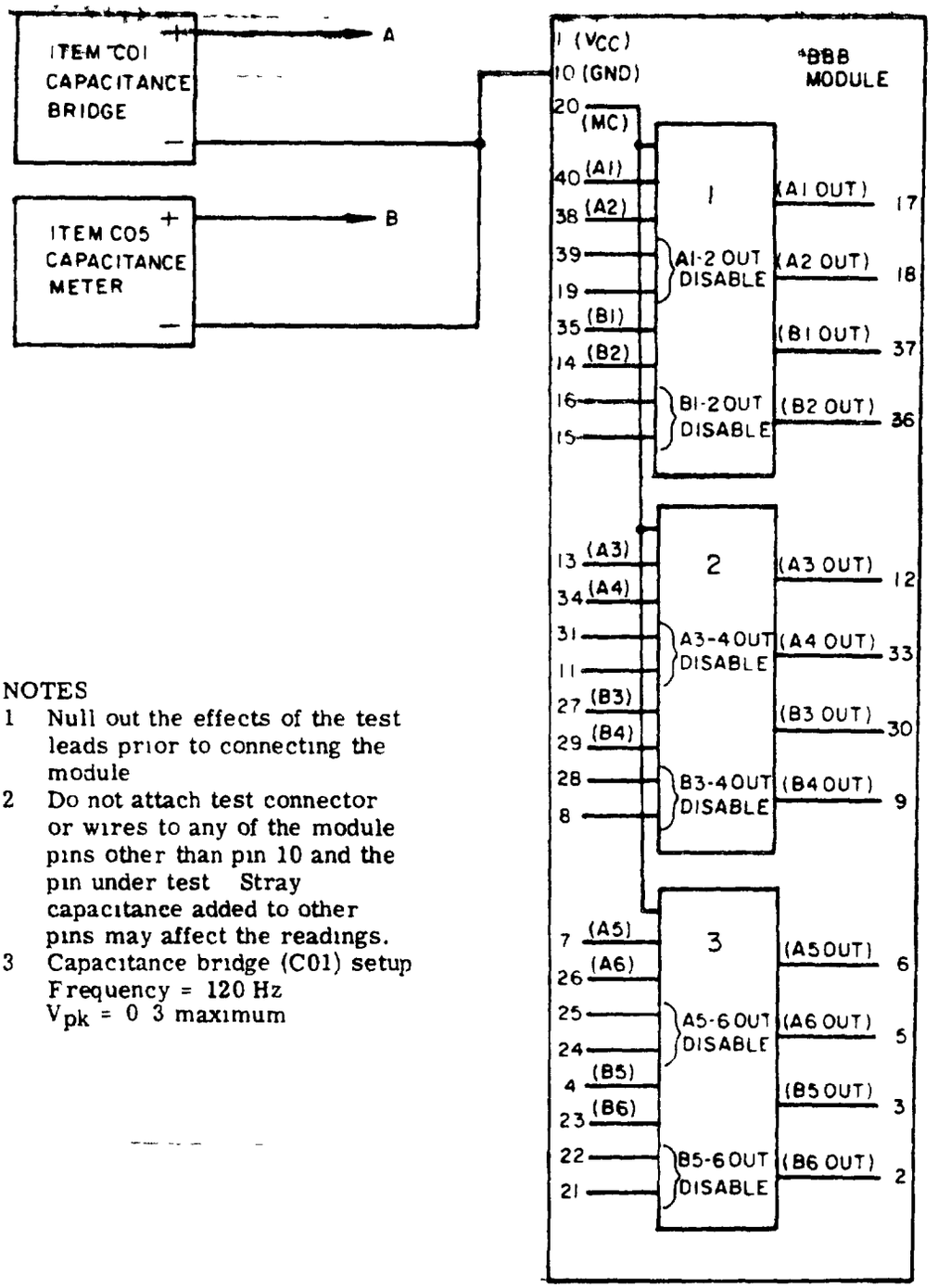
Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
FF18	A6	Input capacitance	Open	26	2	15	pF	32
FF19	B5			4				
FF20	B6			23				
FF21	A5-6 OUT DISABLE			25				
FF22	A5-6 OUT DISABLE			24				
FF23	B5-6 OUT DISABLE			22				
FF24	B5-6 OUT DISABLE			21				
FF25	MC			20	6	35		
GG1	A1 OUT	Output capacitance		17	4	30		
GG2	A2 OUT			18				
GG3	B1 OUT			37				
GG4	B2 OUT			36				
GG5	A3 OUT			12				
GG6	A4 OUT			33				
GG7	B3 OUT			30				
GG8	B4 OUT			9				

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TABLE XXII. Power supply filter, input, and output capacitance test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
GG9	A5 OUT Output capacitance	Open	6	4	30	pF	32
GG10	A6 OUT	↓	5	↓	↓	↓	↓
GG11	B5 OUT		3				
GG12	B6 OUT		2				

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NOTES
1. Null out the effects of the test leads prior to connecting the module
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test Stray capacitance added to other pins may affect the readings.
3. Capacitance bridge (C01) setup
   Frequency = 120 Hz
   V_{pk} = 0 3 maximum

★ FIGURE 32. Power supply filter, input, and output capacitance test circuit for table XXII.

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